Incentives, Evidence, and Reminders for Bureaucrats: Overcoming Barriers to Policy Scale Up*

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Abstract

Scaling effective policies requires the effort of frontline bureaucrats. In a nationwide experiment in the Dominican Republic, we test strategies to increase school principals' implementation of an education program proven effective in a prior RCT. Only 37% of control schools implemented the mandated program, versus 83% in the original trial. Prior adoption experience did not predict compliance, ruling out fixed costs as an explanation. Sharing evidence, providing incentives, and offering assistance had precise null effects. Reminder calls, however, increased implementation by 20 percentage points. A second experiment replicates this finding. Limited bureaucratic attention is thus a critical barrier to policy scale-up.

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1 Introduction

Evidence-based policymaking requires at least three steps: generating evidence on what works, policymakers using such evidence to guide policy choice, and governments successfully implementing the chosen policy. Randomized controlled trials are increasingly used for the first step, while recent work has studied when and how evidence generation affects policy choice (Hjort et al., 2021; DellaVigna et al., 2024; Rao, 2024; García-Hombrados et al., 2026). Yet even when evidence exists and policymakers act on it, execution often depends on frontline bureaucrats with limited bandwidth and many competing demands. Policy selection at higher levels may thus not translate into change on the ground (Muralidharan and Niehaus, 2017; Vivalt, 2020; List, 2022). This paper studies the last step in an otherwise successful example of evidence-based policymaking: what happens *after* an evidence-based policy is chosen by policymakers.

We examine this problem in the context of a nationwide educational intervention in the Dominican Republic. The program, *Aprendiendo el Valor de la Educación* (AVE), provides students with information on the returns to schooling through classroom videos. Building on evidence that correcting misperceptions about education returns can increase schooling (Jensen, 2010), the initial RCT—conducted in two phases in 2015 and 2016 with 1,594 treatment schools—produced positive results: test scores rose by 0.05 standard deviations and dropout rates fell by 3 percentage points (Berry et al., 2018). Encouraged by these results, the Ministry of Education adopted AVE in 2018 and mandated its implementation across all secondary schools.

The experimental and scaled interventions were meant to deliver the same content to students, but their implementation protocols differed. The initial trial relied on intensive in-person workshops to train and motivate school officials. The nationwide rollout, by contrast, used a more standard and less expensive approach to implement the new educational programs. Specifically, the ministry informed school principals about the program by adding AVE to the official school calendar and by sending emails and a follow-up phone call to principals to instruct them to implement it. This shift provides a setting to examine whether implementation frictions that were minimized in the evaluation RCT become binding constraints when delivered as usual

at full scale. We partnered with the government during the rollout to implement experimental interventions designed to overcome these barriers.

Implementation during the national rollout is far from universal. Only 37% of control group schools completed the program and provided proof as ordered by the ministry, compared with a much higher rate of 83% in the original trial. This incomplete implementation could substantially attenuate the program's aggregate impact. It represents an instance of the voltage effect operating through the extensive margin of implementation rather than diminished treatment efficacy (List, 2022).

What explains this implementation gap? We examine several potential mechanisms. First, fixed costs of adoption do not appear to be the main barrier. Implementation is no higher among schools with prior experience—those randomly assigned to treatment in the original RCT. Second, principals do not appear to select into implementation based on perceived student benefits (as measured in a baseline survey of principals). Third, we find suggestive evidence that schools with more managerial overload—particularly where the principal lacks a secretary—show lower implementation rates.

Together, these patterns are difficult to reconcile with a pure cost-benefit explanation and instead point toward organizational or attentional frictions in implementation. We propose a simple framework where principals have limited attention and thus might entirely forget a task when it is time to act. If they do pay attention, they implement the task if its perceived benefits to students and the principal themselves exceed its costs of implementation. The probability of paying attention and the perceived benefits of acting are allowed to depend on the ministry's outreach and motivational efforts towards the principals.

To identify which barriers bind most strongly, we randomized $n = 2{,}724$ schools to different interventions designed to increase implementation. Control school principals received detailed instructions via email and one follow-up phone call, which is the standard government practice. Treatment schools received additional interventions targeting specific frictions: sharing evidence on program effectiveness, providing modest financial incentives, or offering assistance with implementation. We find precise null effects of each of these interventions, consistent with our observational evidence

that neither learning costs nor perceived benefits are the primary constraint. Through the lens of the framework, such interventions that shift perceived costs and benefits are effective only for attentive principals who are close to the margin of implementation. Small effects thus suggest high rates of inattention or few marginal principals.

In contrast, simply reminding principals via two additional phone calls led to a 20 percentage point ($p < 0.001$) increase in implementation as measured by our pre-specified primary outcome. The reminders provided no new information about the program and were more effective among principals with more managerial overload. A second experiment reinforces this finding: when principals were asked to implement a different mandated program (an anti-bullying campaign), reminders again produced large effects (24 percentage points, $p < 0.001$), while messages emphasizing monitoring had statistically significant but smaller effects (4.5 percentage points, $p < 0.01$). The framework implies that interventions that increase attention influence the potentially large mass of infra-marginal but inattentive principals, rationalizing the much larger effects of reminders than treatments targeting costs and benefits. Our evidence is consistent with limited attention or memory as a central constraint on scaling effective policies.

Our findings contribute to three strands of literature. First, they speak to work on the voltage effect—the phenomenon that interventions successful in trials often see smaller effects at scale (Banerjee et al., 2017; List, 2022, 2024). This literature documents several reasons why treatment effects diminish with scale and changes in context (Al-Ubaydli et al., 2017, 2020; DellaVigna and Linos, 2022) and has shown such voltage drops in education interventions when scaled (Bold et al., 2018; Ganimian, 2020; Andersen and Hvidman, 2024). We add evidence of a likely voltage drop in government through incomplete implementation by frontline bureaucrats, with little evidence of favorable selection on expected benefits. We also test several different interventions to reduce this voltage drop.

Second, our findings relate to a large literature on bureaucratic performance and public sector management in the developing world (see Finan et al. (2017); Pepinsky et al. (2017); Besley et al. (2022) for reviews). The primary lens in this literature is one of principal-agent problems. We instead point to a behavioral friction and show that simple reminders can substantially increase compliance with mandated policies, even

when traditional—and more expensive—tools like financial incentives and technical assistance prove ineffective. In doing so, we build a bridge to the behavioral science literature on reminders, which typically target consumers and patients rather than bureaucrats or managers (Thaler and Sunstein, 2021; Szilagyi et al., 2000).

Our findings are remarkably similar to concurrent work by Andersen et al. (2026), who contact schools in Denmark to offer them the use of a free, home-based reading program for students. They find that capturing school leaders' attention through additional emails and phone calls substantially raises adoption of the program, while providing evidence on its benefits has no effect. They study adoption responses to outreach from outside researchers; we instead study implementation of a program mandated by the government after a successful large-scale evaluation. Our findings are also consistent with Gertler et al. (2025), who study behavioral frictions among managers of firms in Mexico, finding that reminders substantially increase the take-up of a profitable offer among small firms. Our results contrast somewhat with those of Andersen and Hvidman (2021), who find that school managers in Denmark responded more to small incentives than to reminders when the Ministry offered them access to an optional program for students.

Third, our results inform research on policy adoption and the use of evidence by decision-makers (Hjort et al., 2021; García-Hombrados et al., 2026; DellaVigna et al., 2024). This work has not separated two key steps: selection of an effective policy and its actual implementation. We focus on a setting where high-level policymakers had already chosen a policy proven effective in an RCT they commissioned, yet on-the-ground adoption remained far from complete. Our null effect of sharing research evidence contrasts with Hjort et al. (2021), who find that evidence increases policy adoption by mayors in Brazil. One interpretation is that different barriers bind at different levels of the bureaucratic hierarchy: mayors may be constrained by lack of information, while school principals face attentional overload rather than informational frictions. Similarly, while DellaVigna et al. (2024) find that organizational inertia predicts adoption of nudge treatments in US cities, we find no effect of prior implementation experience—plausibly because temporary pilot participation does not build the lasting institutional routines needed to overcome such inertia.

The remainder of the paper proceeds as follows. Section 2 develops a conceptual

framework that we use to interpret the treatments and results. Section 3 describes the institutional setting and AVE program. Section 4 presents the experimental design and maps the treatments to the framework. Section 5 outlines the empirical strategy, and Section 6 presents results. Section 7 concludes.

2 Conceptual Framework

To interpret our treatments and the gap between pilot and at-scale implementation, we adapt the scaling framework of Al-Ubaydli et al. (2020), henceforth ALS. We argue that the scale-up wedge in our setting operates through what ALS call *administration quality*—here, the extensive margin of whether schools implement the mandated program at all. We then model the bureaucrat who decides implementation: the school principal. This framework is an ex-post organizing device: we developed it after designing and running the experiment.

2.1 The scale-up problem and the extensive margin of implementation

ALS study how a program's benefit-cost profile changes as it moves from a research study to population-wide implementation, and decompose the resulting wedge into components arising from statistical inference, from the representativeness of the experimental population, and from properties of the situation—spillovers, administration quality, and economies or diseconomies of scale in costs. This framework and the subsequent literature on the science of scaling emphasize the need to take seriously the implementation realities that interventions face at scale (List, 2022, 2024, 2026; Al-Ubaydli et al., 2021). Several components of the ALS framework are limited by design in our setting. The program content is held fixed: the same videos were delivered in the pilot and at scale, so the per-student effect of *receiving* the program, $\bar{T}$, may reasonably be expected to be similar. The two studies also cover largely comparable populations: the pilot randomly sampled approximately 2,500 schools from the public school system nationwide (Berry et al., 2018), while the mandate applied to all secondary schools.[1] The main change is administration quality: in ALS, adherence to

[1]Unlike the pilot, the scale-up sample also includes private schools. However, the results are similar when we restrict the analysis to public schools (Table A.8).

the treatment protocol depends on the per-participant effort devoted to securing it, and declines with scale at constant effort. In our setting, the agents whose adherence must be secured are the roughly 2,700 principals through whom the program reaches students, and their adherence is approximately binary: a school delivers the program or it does not. Our study thus illustrates the case for policy-based evidence made by List (2024): the pilot's intensive workshops obscured an attention constraint that emerged at scale.

Let $q(e)$ be the share of schools implementing when the government exerts outreach and compliance effort e per school. Aggregate net benefits at scale are

$$W(e) \;=\; \underbrace{q(e)\cdot \bar{s}\cdot \bar{T}\cdot N}_{\text{benefits}} \;-\; \underbrace{N\big[\,q(e)\,\kappa \;+\; C(e)\,\big]}_{\text{costs}}, \tag{1}$$

where $\bar{s}$ is the average enrollment, $\bar{T}$ the average treatment effect, N the number of schools, κ the per-school delivery cost conditional on implementing, and $C(e)$ the per-school cost of the outreach technology.

The voltage drop, in our setting, comes from a reduction in the government compliance effort e, such that q fell from 0.83 to 0.37—the voltage effect operating through the extensive margin of implementation. We assume that, if a school complies, the treatment effects on students, $\bar{T}$, remain unchanged.[2] The policy problem for the government is to choose an outreach technology, and the relevant statistic for comparing technologies is the cost per marginal implementing school, $\Delta C/\Delta q$, which we estimate in Section 6. ALS treat the share complying, $q(e)$, as a technological object; to unpack it, we model the principal who decides implementation.

2.2 The principal's problem

We assume that implementation of a mandated task requires two things of principal i. First, the task must come to mind at a moment when action is feasible; we call this *attention* and denote its probability $a_i \in [0,1]$. Principals manage many competing responsibilities, and a task that is never salient at an actionable moment

[2]The assumption of an unchanged treatment effect conditional on adoption is a simplification, but a plausible one given that we do not find substantial selection into implementation based on perceived or predicted treatment effects.

goes undone regardless of its merits. Second, conditional on attending, the principal implements if net benefits are positive:

$$v_i \;=\; b_i^S + b_i^P - c_i, \tag{2}$$

where b_i^S is the perceived social benefit (gains to the principal's students, weighted by intrinsic motivation), b_i^P the private benefit (expected sanctions from the Ministry avoided by complying, plus any explicit rewards), and c_i the hassle cost of implementation, which may include a fixed learning component that is sunk for schools that implemented the pilot. With an idiosyncratic taste shock ε_i drawn from a continuous distribution, the probability that principal i implements is

$$\pi_i \;=\; a_i \cdot G(v_i), \qquad q = \mathbb{E}\big[a_i\, G(v_i)\big], \tag{3}$$

where $G(v) \equiv \Pr(\varepsilon_i > -v)$ is increasing with density g.[3]

Comparative statics. The two margins have different comparative statics:

$$\frac{\partial \pi_i}{\partial a_i} = G(v_i), \qquad\qquad \frac{\partial \pi_i}{\partial v_i} = a_i\, g(v_i). \tag{4}$$

An attention intervention moves *every* principal whose net benefit is already positive but who would otherwise have failed to act: its effect is scaled by the level $G(v)$. A benefit- or cost-side intervention moves only principals who are both attentive *and* close to the margin: its effect is scaled by $a\, g(v)$. If most attentive principals are well inside the implementation region—$G(\bar{v})$ high, $g(\bar{v})$ small—then even sizeable shifts in benefits or costs produce small effects, while restoring attention produces large ones.

Attention technology. Attention is produced by the government's contacts. Suppose contact k occurs at time t_k with salience $\rho_k \in [0,1]$ that depends on the medium (e.g., a phone call is more salient than an email), and places the task in the principal's mind during the action window w with probability $\rho_k\, d(w - t_k)$, where $d(\cdot) \in [0,1]$ decreases in the delay between contact and window, with $d(0) = 1$. If contacts operate independently,

$$a_i \;=\; 1 - \prod_{k \in \mathcal{K}} \big[\, 1 - \rho_k\, d(w - t_k) \,\big], \tag{5}$$

[3]The separation of a memory stage from a cost-benefit stage follows models of limited memory and reminders (Karlan et al., 2016; Ericson, 2017).

so the marginal contribution of contact k is $\rho_k\, d(w - t_k) \cdot (1 - a_{-k})$, where a_{-k} is the attention generated by the other contacts. Contacts close to the action window and delivered through salient media contribute the most: a well-timed, salient fifth contact may add more than four earlier, low-salience ones combined.[4]

Contacts may do more than jog memory: they may signal information about the program's efficacy (shifting b^S) or about the Ministry's priorities and monitoring intent (shifting b^P). The total effect of an additional contact is, therefore,

$$\Delta\pi_i \;=\; \underbrace{G(v_i)\,\Delta a_i}_{\text{attention}} \;+\; \underbrace{a_i\, g(v_i)\big[\Delta b_i^S + \Delta b_i^P\big]}_{\text{signaling}}. \tag{6}$$

The signaling term takes the same form as interventions that shift net benefits directly. The effects of those direct interventions thus provide some guidance regarding its plausible contribution to any reminder effect (Section 6).

Implications. The framework has three implications that organize our results. **I1** (*levers*): if the distribution of net benefits has little mass near the margin while most principals have positive net benefits, attention interventions produce large effects and benefit- or cost-side interventions small ones. **I2** (*heterogeneity*): the attention gain from an added contact, $\rho\, d \cdot (1 - a_{-k})$, decreases in baseline attention, so attention interventions should be most effective for principals with the least attentional slack (e.g., the most managerial overload); and if variation in a_i dominates variation in v_i, implementation will be weakly correlated with perceived benefits in the cross-section. **I3** (*an upper bound*): attention interventions cannot push implementation above the share of principals with positive net benefits, $q \leq \mathbb{E}[G(v_i)]$; principals with $v_i < 0$ require interventions that increase net benefits, not more reminders.

3 Institutional Setting and the AVE Program

The education system in the Dominican Republic consists of six years each of primary and secondary education. Of all secondary schools, 74% are public. Public secondary schools are run by principals appointed by the Ministry of Education. Principals have

[4]Evidence from other settings is consistent with both features: interactive phone contact outperforms one-way text messages (Angrist et al., 2022, 2023), and more messages are not always better (Cortes et al., 2021; Doss et al., 2019).

wide-ranging authority within the school but are required to follow the official school curriculum designed by the Ministry of Education. Schools are organized into 122 educational districts.

School quality is low, and dropout is widespread. The Dominican Republic ranked last in international student performance assessments in 2013 and 2015 (UNESCO, 2015; OECD, 2016), and approximately 38% of students who take the 8th-grade national exam do not complete high school.

Large-scale 'pilot' evaluation. Motivated by earlier research that shows that students underestimate the returns to schooling (Jensen, 2010), the Ministry of Education collaborated with academics to create a scalable version of an education information campaign. This partnership produced the "Aprendiendo el Valor de la Educación" (AVE) program, which comprises four 15-minute videos designed to provide secondary school students with information about the potential monetary and non-monetary benefits of education. Earnings information was drawn from nationally representative surveys and presented in a comprehensible manner. For instance, wages are discussed by showing the distribution of wages for 100 representative adults for a given level of schooling. The videos also illustrated how factors such as luck and student performance contribute to outcomes. Implementation involved students watching the videos during class and discussing them with their teachers.

The program was evaluated in a randomized field experiment conducted in 2015 and 2016, as outlined by Berry et al. (2018). In the initial round of the study in 2015, the Ministry of Education randomly assigned 398 public schools to the treatment group to implement the program for 7th and 8th-grade students and 200 public schools to the control group.[5] School officials from all treated institutions were required to attend a two-day workshop in Santo Domingo, during which they received copies of the videos and detailed instructions for implementation. Subsequently, schools received multiple follow-up phone calls to ensure program adherence.

In the second round of the study in 2016, the program expanded to include stu-

[5]The initial experiment consisted of two treatment arms in which the program either emphasized quantitative statistics or qualitative information. Since both treatment arms had similar effects on test scores and dropout, we pool them together. The scale-up version of the program was based on the quantitative statistics treatment.

dents across all secondary school grades (7th through 12th). The sample comprised 2,469 public schools, of which 1,594 were assigned to the treatment group. Schools from the first round maintained their original treatment assignments. Due to the larger scale, the program was implemented differently. The Ministry of Education initially provided the program material to the educational districts, which then organized in-person workshops with local schools to facilitate program implementation. Implementation rates—achieved through intensive in-person engagement—were 97% in the first round and 80% in the second, making for an average implementation rate of 83%. Intent-to-treat estimates from Berry et al. (2018) reveal that the AVE program increased test scores by 0.05 standard deviations and reduced dropout rates in the following year by 3 percentage points.

Nationwide adoption and scaling. Based on these findings, the Ministry of Education decided to expand the AVE program in the 2018-19 academic year.[6] Under the new initiative, every student enrolled in 7th grade or above was required to watch the program's videos within the school setting. To facilitate this, the government designated September 24–28 as AVE week, and incorporated the intervention into the official school calendar (see Online Figure B.1).

To scale the program, the government further modified the implementation approach. Instead of conducting in-person workshops with principals, the scaled-up version of the program instructed principals to download the necessary materials, including the videos and implementation protocols, from an online platform. These instructions and protocols were communicated by the Ministry to principals via phone and email.

The school calendar—and thus the AVE program—is in theory compulsory for all public and private schools. The emails and phone calls the Ministry made to principals referenced the school calendar, making the mandatory nature of the program clear to principals. In the baseline survey, described below, 90% of control group principals reported that a hypothetical school principal who failed to comply with Ministry of Education directives would likely face sanctions. In practice, however, enforcement is

[6]The government did not require any schools to implement the AVE program during the 2017–18 academic year, resulting in a one-year gap between the second round of the initial study and the nationwide scale-up.

weak. To our knowledge, schools that did not implement the AVE program faced no direct penalties for noncompliance.

4 Experimental design

After the Ministry decided to scale up the program, we partnered with them to experimentally test which interventions could increase implementation by school principals. Implementation rates in the original trial were 83%, suggesting at least some room for improvement. Moreover, we anticipated that switching from in-person workshops to the usual channels for informing and motivating principals might further reduce implementation.

4.1 Sample selection

Our initial sampling frame was all 4,351 schools which offered secondary school grades. We excluded schools with fewer than five students enrolled in secondary school grades in 2017 (n=162). We further restricted the sample to the 3,888 schools that participated in either the 8th-grade National Exam in 2016 or the 12th-grade National Exam in 2018, effectively excluding informal schools from our analysis (n=301).

We dropped schools without contact details (n=7), and in cases where principals supervised multiple schools meeting our sample criteria, one school was randomly selected, and the others were excluded (n=119). Among the remaining 3,762 schools, we dropped those whose principals were unreachable by phone during the baseline survey (n=626), those where the school principal did not have email access (n=343), a few schools that were used for piloting the surveys (n=44), and a small number of additional administrative exclusions (n=25).[7] Ultimately, our final sample consisted of 2,724 principals. Table A.1 shows a detailed breakdown of the sample selection process. The differences between the experimental sample and the universe of secondary schools in the Dominican Republic are quite small, although often statistically significant due to large sample sizes (Table A.2). Two relevant differences are that

[7]The sample for the original AVE experiment was selected without conditioning on baseline survey completion. If principals who are uncontactable for surveys are also less likely to implement the program and report compliance, our measured implementation rates in the scale up might even over-estimate compliance at scale relative to the original AVE study.

our final sample has slightly larger schools than average (212 versus 183 students) and principals are slightly less likely to have no secretary (24% versus 28%).[8]

4.2 Implementation protocol

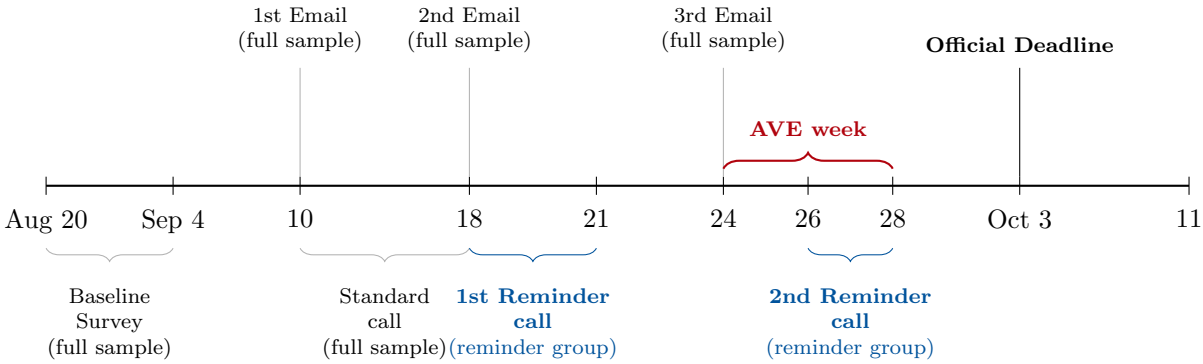
The academic term began on August 20. Between August 20 and September 4, we attempted to call 3,762 principals to conduct baseline surveys over the phone, successfully reaching 83% of them.

The academic calendar, posted by the Ministry before term began, indicated to all schools that AVE week was September 24 to 28. On September 10—two weeks before AVE week—the Ministry emailed all school principals a link to the AVE platform, from which school staff could download the program's videos and implementation protocol, along with a brochure with instructions (see Figure B.2 for an example). During the following week, the Ministry called all principals to repeat this information and direct attention to the email. On September 18 and September 24—one week before and the start date of the AVE week—principals were sent a second and third email repeating these instructions. All emails were sent from official Ministry of Education email accounts and calls were made by public employees at an established call center under the Ministry of Education, which had supported several Ministry initiatives. The Ministry instructed principals to submit a form and a photo of students watching the AVE videos as proof of implementation, with a deadline of October 3. Figure 1 shows the full timeline.[9]

[8] Although differences in observable characteristics are relatively small, the exclusion criteria may still affect external validity. Excluding schools with very few students is unlikely to be consequential, but principals who could not be contacted or who supervised multiple schools may have been less likely to implement the program on average. Whether they would have higher or lower treatment effects is ambiguous, but it seems plausible that they would react less to the reminder intervention, which involved two additional phone calls. On the other hand, they may also be less likely to implement the program to begin with, leaving more room for gains due to policy interventions.

[9] As of October 11, only 41% of principals had submitted a photo. Reacting to this low compliance, the Ministry then sent two additional emails and made two follow-up calls to all principals who had not yet responded. We use October 11 as the cut-off date for our analysis and also report effects as of December 1 as a robustness check.

Figure 1: Implementation Timeline



Notes: This figure shows the timeline of communications sent to principals. Items labeled "(Full Sample)" were sent to all principals. Items labeled "(Reminder Group)" were sent only to the reminders treatment group. All other treatments began on September 10, when the first email was sent to the full sample.

4.3 Treatment arms

We randomized schools into several treatment arms, each targeting a different potential barrier to implementation: "Evidence" (sharing research findings), "Incentive" (financial motivation), "Reminders" (additional follow-up calls), and "Assistance" (implementation support). We describe each below and provide the full protocols and scripts in Appendix C.

Evidence Treatment. In the first treatment arm, principals received information about the results of the AVE pilot. Specifically, the emails and phone call script included the following paragraph:

> *It is important to note that a research study, carried out between 2014 and 2016 in 2,500 schools in the Dominican Republic, showed that implementing this campaign significantly improves scores on national tests and decreases dropout rates. The schools that showed the videos increased national test scores by 0.6 points in total. This study shows that if all the schools in the country showed these videos to their students, the number of dropouts in secondary schools would decrease by 6,500 students per year.*

This information was also conveyed through easy-to-read figures in the brochure

sent to principals in the evidence group (see Figure B.3).

Incentive Treatment. In a second treatment arm, principals received a financial incentive in the form of a lottery. The emails and the call scripts included this extra paragraph:

> *Please note that you have been randomly selected to be one of the directors who can participate in the draw for a Samsung tablet to help you run the school. You will enter this lottery only if you implement the AVE campaign at your school as outlined in the instructions below in this email. One out of 10 people who enter the drawing will win a tablet.*

The value of the tablets was equal to 16% of the average monthly salary of a school principal. This treatment was expected to serve as a monetary benchmark for the effectiveness of other treatment arms.

Reminders Treatment. In a third treatment arm, principals received two additional reminder calls from the Ministry. The first reminder call was between September 18 and 21, after the standard calls to every school in the sample were completed, and about one week before AVE was to be implemented. The second reminder call was placed during the AVE week. Calls were made by public employees at the Ministry's call center, just as in the first call that was made to all schools in the study.

Assistance Treatment. A fourth treatment arm offered principals additional assistance through a helpline and a link to a website with further resources. This was meant to provide help if schools had any technical difficulties with downloading or screening the videos.

Randomization was conducted at the school level and stratified by school district. We cross-randomized the four treatments, with the exception that we excluded the group that contained both reminders and incentives to preserve budget and because we (incorrectly) anticipated high implementation rates and potentially ceiling effects. Figure A.1 illustrates the treatment cells and Table A.10 reports the minimum detectable effect sizes for each possible comparison.

4.4 Follow-up Experiment

We also conducted a follow-up experiment in which the same principals were asked to implement an anti-bullying campaign through a similar process. Like the AVE campaign, the anti-bullying campaign appeared on the official school calendar and was therefore mandatory for all schools. We experimentally varied whether principals received two reminder calls, were told that their compliance would be monitored, and received a motivational message (see Appendices D and E for details). Unlike in the first experiment, control group schools received only one email and no phone calls.

4.5 Mapping treatments to the framework

Table 1 maps each treatment arm, and the key sources of observational variation, to the parameters of the framework in Section 2. The evidence treatment shifts perceived student benefits b^S; the incentive treatment shifts private benefits b^P, though its expected value is modest (a one-in-ten lottery for a tablet worth 16% of a month's salary, i.e. under 2% of monthly salary in expectation); and the assistance treatment lowers the hassle cost c. All three operate through the net-benefit margin $a\, g(v)\, \Delta v$ in Equation (4). The reminders treatment, by contrast, adds two salient, well-timed contacts—one the week before and one during AVE week—and thus operates primarily on attention a, though it may also carry a signal of Ministry priority. In the second experiment, the motivational and monitoring messages shift b^S and b^P respectively, while the reminder arm again targets a. Because benefit-side treatments operate only through attentive principals, their effects are scaled by a and should be weakly larger when cross-randomized with reminders, though our power for these interactions is limited.

Through the lens of the framework, the pilot and scale-up differ in one critical way: the attention technology changed. The pilot's two-day in-person workshops make the task effectively impossible to forget—setting $a \approx 1$. The scaled rollout's standard protocol likely delivers much less attention.

The Ministry's decision to adopt AVE nationwide might also have signaled the program's efficacy and political priority to school principals. Indeed, 83% of control group principals in the baseline survey stated that they believed an intervention like

Table 1: Treatments and the Parameters They Target

Intervention	Parameter	Effect operates through	Source
Reminders (2 extra calls)	$a \uparrow$ (and possibly b^P)	$G(v)\,\Delta a\,[+\,a\,g(v)\Delta b^P]$	Experiments 1 & 2
Evidence	$b^S \uparrow$	$a\,g(v)\,\Delta b^S$	Experiment 1
Incentive (tablet lottery)	$b^P \uparrow$	$a\,g(v)\,\Delta b^P$	Experiment 1
Assistance (helpline)	$c \downarrow$	$a\,g(v)\,\Delta c$	Experiment 1
Monitoring message	$b^P \uparrow$	$a\,g(v)\,\Delta b^P$	Experiment 2
Motivational message	$b^S \uparrow$	$a\,g(v)\,\Delta b^S$	Experiment 2
Pilot-treatment status	$c \downarrow$ (sunk fixed cost)	$a\,g(v)\,\Delta c$	Prior experiment
Perceived benefits (survey)	b^S, b^P	cross-sectional variation in v	Observational
Managerial overload (secretary, roles)	a	cross-sectional variation in a	Observational

Notes: This table maps each experimental intervention, and the key sources of observational variation, to the parameters of the framework in Section 2: the attention probability a, perceived social benefits b^S, private benefits b^P, and implementation/hassle costs c.

AVE would significantly improve student outcomes, and 90% believed non-compliance with Ministry instructions would likely be sanctioned. In the model's terms, the scale-up control group is thus characterized by high v and low a.

4.6 Summary statistics and balance tests

Table A.3 shows summary statistics and balance across treatment arms, using both administrative data and the baseline survey. Panel A reports the characteristics of schools and Panel B reports characteristics of principals. In our control group sample, 80% of schools were public and 74% were located in urban areas. Most schools had the necessary infrastructure to easily implement the AVE campaign, including electricity (99%), a projector (76%), and an internet connection (73%).[10] One-third of the sample was part of the treatment group in the previous pilot study of the AVE program. In the baseline survey, about half of principals say that students undervalue education and 83% say that providing information on the benefits of completing school would significantly improve test scores and decrease dropout rates. Treatment and control groups are largely balanced across school and principal characteristics.

[10] If a school did not have an internet connection, principals could still download the AVE material from a cybercafe or at home.

5 Outcome measures and empirical strategy

Our outcome of interest—implementation of the AVE program—is measured using different types of administrative and student data. To study heterogeneity of treatment effects, we also leverage the baseline survey and administrative data.

Emails were sent through a marketing platform that tracked whether each principal received, opened, and clicked on the email. The email contained links to an online platform where principals registered to download the AVE campaign material; we observe downloads linked to each principal. We recorded all phone calls made to the schools. Overall, 98% of principals received the initial email and 75% opened it (Table A.4, columns 1 and 2).[11] 95% of principals answered the first call from the Ministry, which repeated the information provided in the email (column 7). These rates are similar across treatment arms.

Principals were instructed to submit a photo via WhatsApp and a form with implementation details via email. Following our pre-analysis plan, our primary outcome is an indicator for whether the principal submitted a photo of students watching the AVE videos, which the Ministry required as proof of implementation. We also examine download rates and form submissions as secondary measures of implementation. We also aggregate information on downloads, photo and form submissions using a standardized index per Anderson (2008). We supplement these outcomes with information gathered through student questionnaires. Specifically, for students who took the 8th-grade or 12th-grade national exam in 2019, we included questions in the national exam asking whether their school had participated in the AVE campaign during the 2018-2019 academic year. This provides an entirely independent measure of implementation, with the caveat that it measured recall a year later for only a subset of students, and schools in all arms received additional reminders after our main study period ended (based on our results).

The primary treatment effects of interest are estimated by calculating the intent-

[11] Since the treatment information was also repeated in the calls, email opening rates are endogenous.

to-treat (ITT) estimate of take-up via OLS:

$$Y_i = \sum_k \beta^k \cdot T_i^k + X_i'\gamma + \epsilon_i \tag{7}$$

where Y denotes the outcome for principal i, T^k is an indicator for assignment into the treatment arm k, and coefficients β^k are the parameters of interest. Our main analysis focuses on the average impact of each of the four treatments (without treatment interactions), but we also report the effects for each treatment cell. The vector X_i includes stratification dummies and baseline covariates selected via the double lasso approach by Belloni et al. (2014) from the covariates listed in Table A.3. Table A.5 reports q-values corrected for multiple hypotheses testing.

Preregistration and deviations from pre-analysis plan. Our experiment was pre-registered on the AEA RCT Registry. We follow the pre-analysis plan with minor deviations, detailed in Table A.6. In the text, we mention when an analysis was exploratory (i.e., not pre-registered).

6 Results

6.1 Descriptive results

We begin by examining program adoption in the control group. Overall implementation rates are modest: only 37% of control schools submitted photographic proof of implementation and 27% submitted the form, compared with implementation rates—measured similarly—of 83% on average in the original trial.[12] Indeed, only 69% of principals even downloaded the material to begin with.[13] This gap illustrates the "voltage effect" operating through the extensive margin (List, 2022): while the

[12]The closest benchmarks for this 37% rate are two other at-scale school mandates in the Dominican Republic. In our second experiment, 18% of control schools submitted a photo documenting implementation of a mandated anti-bullying campaign, which also appeared on the official school calendar. In Morales et al. (2026), 11% of control schools submitted photos of a mandated workshop on national assessment results, and 5% submitted a school improvement plan based on those results. The rate in our main experiment is higher than in both those cases, although our control schools, unlike theirs, received a phone call.

[13]Among schools which had not been part of the treatment group in the previous trial two years ago, and thus had not had any access to the AVE materials whatsoever, download rates were still only 70%.

Table 2: Predictors of AVE Campaign Adoption

		Primary Outcome: Submitted Photo			
	Mean of Predictor in the Control Group	Only Control Group		Full Sample	
		Coeff.	N	Coeff.	N
	(1)	(2)	(3)	(4)	(5)
Managerial Overload Index	0.507	-0.076	302	-0.064**	2,058
		(0.070)		(0.028)	
Principal Has Multiple Positions	0.656	-0.098	294	-0.031	1,994
		(0.061)		(0.024)	
Principal Has No Secretary	0.264	-0.067	220	-0.103***	1,499
		(0.073)		(0.029)	
Benefits Index	0.682	0.005	380	0.048	2,724
		(0.076)		(0.031)	
Believes Extra Info Would Have an Effect on Students	0.826	0.049	380	0.007	2,724
		(0.064)		(0.027)	
Believes Penalty after Non-Compliance is Very Likely	0.537	-0.024	380	0.032*	2,723
		(0.050)		(0.019)	
Fixed Costs of Implementation					
School Was in the Treatment Group in Pilot	0.313	-0.042	380	0.030	2,724
		(0.053)		(0.020)	
Outcome Mean		0.374		0.407	

Notes: This table examines predictors of AVE campaign adoption. Column (1) shows the mean of the predictor in the control group. In Columns (2) and (3), we regress a dummy for whether the principal submitted a photo of the AVE campaign adoption on the corresponding predictor. Column (2) is restricted to the control group and Column (3) includes the full sample. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

program to be delivered to students is itself unchanged, the lighter implementation protocol at scale sees many schools fail to deliver the program.

Table 2 examines predictors of program adoption. In an exploratory analysis, we group baseline characteristics into three classes of factors: (i) managerial overload, (ii) perceived benefits of implementation, and (iii) fixed costs of implementation. The managerial overload index proxies for bandwidth constraints and uses information from government payroll data. It averages two indicators: whether the principal has multiple positions (typically serving as a teacher in addition to being the principal) and whether the principal lacks a secretary. A secretary handles correspondence and helps manage the principal's schedule, among other tasks. The benefits index averages

principals' baseline survey responses regarding likely student benefits from AVE and likely penalties for non-implementation. Variation in fixed costs of implementation is captured by whether the school was in the treatment group in the AVE pilot. The idea is that, having previously implemented the program, any fixed costs of learning how to implement the program should already be sunk.[14]

Strikingly, principals who perceive the program to be beneficial to their students are no more likely to actually implement the program in the scale-up, either in the control group or in the full sample. Schools do not appear to select into program implementation based on (perceived) benefits, as one might hope in case of incomplete adoption. Nor does past experience with having implemented the program affect compliance.[15] In contrast, we do find suggestive evidence that principals with less managerial overload and thus greater attentional bandwidth are more likely to implement the program. This pattern is consistent with Implication I2 of the framework: when variation in attention a, rather than in net benefits v, is the dominant source of heterogeneity, implementation should be correlated with proxies for attention but may be only weakly correlated with perceived benefits.

6.2 Experimental results

Table 3 reports treatment effects on program adoption. Our prespecified primary outcome is whether the principal submitted a photo of students participating in the AVE program (column 1). Secondary outcomes include whether the principal downloaded the campaign material (column 2) and submitted the mandatory form (column 3). Column 4 reports a standardized adoption index combining these three

[14]The program is a quite simple one—downloading and screening videos on a projector and facilitating a discussion with students after they watch the videos. Still, given limited technical capacity and a high workload, this may not have been a trivial cost for some schools. Prior participation could also reduce adoption if principals believed that repeating the program was unnecessary. We consider this channel less likely, however, because new cohorts of students who did not participate in the earlier AVE program would still benefit from it.

[15]This result holds when restricting the sample to schools that were part of the pilot RCT. Thus, we find no causal effect of past implementation on implementation rates in the scale-up (Table A.7). This contrasts with DellaVigna et al. (2024), who find that organizational inertia predicts adoption of nudge interventions in U.S. cities. One possible explanation for the difference is that temporary participation in a pilot may not be sufficient to establish persistent routines, or that the one-year gap between the initial experiment and the scale-up was enough to erode any institutional learning.

measures.[16]

Table 3: Treatment Effects on AVE Campaign Adoption

	Index Components			
	Primary Outcome	Secondary Outcomes		
	Submitted Photo	Downloaded Material	Submitted Form	Adoption Index (in Std)
	(1)	(2)	(3)	(4)
Reminders	0.203***	0.150***	0.131***	0.411***
	(0.022)	(0.019)	(0.021)	(0.042)
Evidence	-0.011	-0.009	-0.030*	-0.043
	(0.019)	(0.018)	(0.018)	(0.039)
Incentive	0.010	-0.023	0.055**	0.031
	(0.024)	(0.024)	(0.023)	(0.051)
Assistance	-0.012	-0.035**	-0.020	-0.062*
	(0.018)	(0.017)	(0.017)	(0.036)
Mean of Control Group	0.374	0.687	0.271	0.000
Observations	2724	2724	2724	2724

Notes: This table shows the treatment effects on different adoption measures of the AVE campaign. We regress each outcome on indicator variables for assignments to the evidence treatment group, the incentive treatment group, the reminders treatment group and the assistance treatment group, stratification dummies, and baseline controls selected by LASSO. Table A.3 shows the list of potential lasso controls. The adoption index (column 4) is standardized to have mean zero and standard deviation one in the control group. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Across most outcomes, we observe precise null effects for the evidence, incentive, and assistance treatments. Sharing research findings on the program's effectiveness does not increase adoption, nor does offering a financial incentive through a lottery or providing implementation support. These null results are consistent with the descriptive patterns in Table 2. Principals who perceive the program as more beneficial are

[16]Table A.8 shows that we find similar results if we exclude private schools or schools that were part of the treatment group in the initial pilot study.

no more likely to implement it; nor does experimentally providing evidence on benefits increase implementation. Similarly, the null effect of offering technical assistance with implementation is consistent with previous experience having no effect on implementation in the nationwide scale up. These nulls are jointly consistent with Implication I1 of the framework: all three treatments operate on the net-benefit margin—b^S, b^P, and c, respectively—so their effects are scaled by the mass of attentive principals near indifference, $a\,g(\bar{v})$. Consistent with most principals already reporting believing the program was beneficial and sanctions likely, that mass appears to be small. The nulls do not imply that benefits are irrelevant, only that few principals are at the margin of indifference; substantially larger incentives or sanctions could still matter.

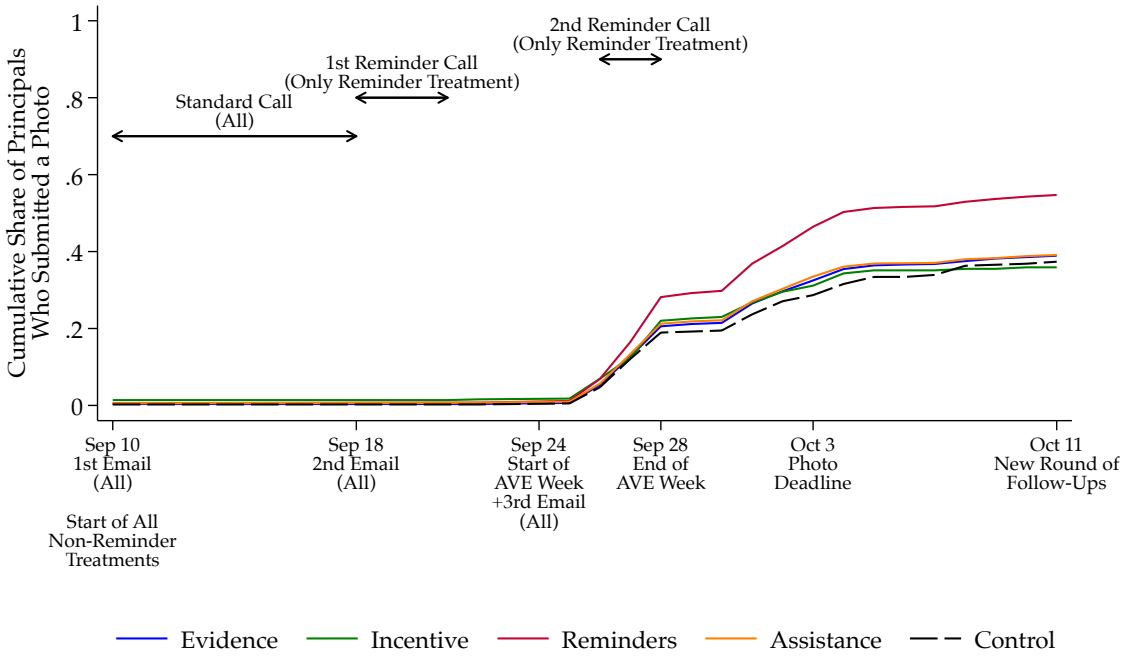
The only intervention that increased program adoption is the reminders treatment. Principals who received two additional follow-up calls showed a 20-percentage-point increase in photo submission (column 1), a 54% increase relative to the control group mean. This effect is large and highly significant ($p < 0.001$), and holds across all outcome definitions. This finding remains highly statistically significant ($q = 0.001$) when controlling the False Discovery Rate using the method of Benjamini et al. (2006) (Table A.5).

When breaking the results up by each treatment cell in the cross-randomized design, treatment effects for the primary outcome are similar and statistically significant across each cell which involves a reminder, while all other treatment cells remain non-significant (Table A.9). While our framework predicts that combining reminders with interventions that further increase principals' perceived net benefits from implementation should raise adoption, we do not detect statistically significant incremental effects of either the evidence or assistance treatment beyond those of reminders alone (Table A.10). However, we have limited statistical power for some of these comparisons: Table A.10 reports ex-post minimum detectable effect sizes (MDEs) of 9 to 17 percentage points when comparing fully disaggregated treatment cells with each other, as opposed to MDEs of 5 to 7 percentage points for our preferred pooled comparisons.

Figure 2 plots implementation over time. The cumulative adoption curves for the evidence, incentive, assistance, and control groups are nearly indistinguishable throughout the entire sample period, reinforcing that the null effects are not a matter

of delayed response. The reminders group, by contrast, shows persistently higher adoption, with the gap emerging during the AVE week and widening through the deadline.[17]

Figure 2: Take-Up Rate Over Time by Treatment



Notes: This figure shows the cumulative share of principals who submitted a photo of the implementation of the AVE campaign in their school over time. The blue line corresponds to the evidence treatment, the green line to the incentive treatment, the red line to the reminder treatment, the orange line to the assistance treatment, and the black dashed line to the control group. The kink between September 29 and 30 corresponds to the weekend.

Implementation versus reporting. An important question is whether the reminder treatment affected the actual implementation of AVE in schools or merely its *reporting*—either truthful or exaggerated—to the Ministry. For example, Muralidharan and Singh (2020) find that when school officials in India were tasked

[17]While photo submissions only respond around the second reminder, downloads—which were feasible before AVE week—respond already to the first reminder call (Figure A.2).

with implementing a school governance reform, they created an illusion of compliance by completing the required paperwork without making meaningful changes in their schools. In our setting, several pieces of evidence point towards an increase in implementation itself, and not merely in reporting.

First, the reminder treatment had similarly sized effects on *downloads* of the AVE program videos, which were necessary for implementation. Second, in collaboration with the Ministry of Education, we included questions in the national examinations administered approximately one year after our experiment asking students whether they or their peers had participated in the AVE program. Students in reminder-group schools were significantly more likely to recall the AVE videos, although the estimated effects of 3 to 6 percentage points were smaller than those based on our main administrative measures of implementation (Table A.11). In contrast, the incentive treatment increased submission of the implementation form but did not increase student recall. We interpret this as evidence that reminders moved real implementation, while incentives moved paperwork.

Several factors may explain why the estimated effects on student reports are attenuated. First, the student survey was administered nearly one year after the intervention. Moreover, after learning about our results, the Ministry sent additional follow-up emails and called principals who had not submitted evidence of implementation by the original deadline. These subsequent efforts likely narrowed the difference between the treatment and control groups.[18] Second, some students may have paid limited attention to the question, forgotten the videos, confused them with those shown during the AVE pilot, or failed to recognize the program by its official name. Third, the student reports come from only 58% of the schools in the experiment, since not all schools had students taking the national exam that year. The treatment effect on photo submissions was smaller in these schools to begin with (17 versus 20 percentage points, Table A.13).

Still, the student reports appear to capture a meaningful signal and are systematically related to our primary outcome measures. Figure A.3 plots the distribution

[18]Consistent with this interpretation, Table A.12 shows that the effects of the reminder treatment on our administrative measures of adoption persist but become smaller when we extend the measurement period to December 1, by which time the government had conducted additional follow-ups with all schools in the sample.

of the share of students reporting that they or their peers participated in the program, separately for schools classified as adopters and non-adopters under each of our three administrative measures. Across all three measures, the distribution of student-reported participation differs markedly between the two groups. For example, the average share of students reporting participation in AVE Week is 63% among schools that submitted a photo, compared with 41% among schools that did not. Nevertheless, some students reported seeing the videos even in schools where the principal had not downloaded them, suggesting some degree of misreporting or imperfect recall.

Finally, Table A.13 reports results using different combinations of the outcome measures, providing upper and lower bounds based on more or less stringent definitions of adoption. Under a definition that requires schools to download the materials and submit either a photo or the implementation form, the reminder treatment increases adoption by 21 percentage points. Under the stricter definition requiring downloads and the submission of both the form and a photo, the estimated effect is 11 percentage points. When student reports are incorporated into the adoption measures, the estimated effects of the reminder treatment range from 8 to 15 percentage points. Thus, across a range of definitions, the reminder treatment consistently produces substantial increases in AVE adoption, reassuring us that it affected actual program implementation.

Follow-up experiment. In Table 4, we report results from a follow-up experiment in which the same principals were required to implement an anti-bullying campaign through a similar process. Reminders again produced large effects, increasing adoption by 24 percentage points ($p < 0.001$). The monitoring message increased take-up by 4.5 percentage points ($p < 0.01$).[19] The motivational message emphasizing the importance of the anti-bullying campaign had no effect on adoption, mirroring the null effect of the evidence treatment in the AVE experiment. The pattern across both experiments is consistent: principals respond strongly to reminders of new tasks, while interventions that pull more traditional levers—providing evidence or motivational messages, financial incentives, implementation assistance, or monitoring—have much smaller effects.

[19]This message informed the principal about the observed compliance with the AVE program, and stated that "We will be monitoring compliance with this instruction, to ensure that this campaign is implemented properly."

Table 4: Treatment Effects on Anti-Bullying Campaign Adoption

	Index Components			
	Primary Outcome	Secondary Outcomes		
	Submitted Photo	Clicked on Declaration	Submitted Form	Adoption Index (in Std)
	(1)	(2)	(3)	(4)
Monitoring	0.045***	0.035**	0.062***	0.126***
	(0.017)	(0.017)	(0.017)	(0.036)
Motivation	-0.011	-0.018	-0.004	-0.030
	(0.017)	(0.017)	(0.017)	(0.036)
Reminders	0.241***	0.281***	0.178***	0.638***
	(0.017)	(0.018)	(0.018)	(0.038)
Mean of Control Group	0.182	0.379	0.215	0.000
Observations	2724	2724	2724	2724

Notes: This table shows the treatment effects on different adoption measures of the anti-bullying campaign. We regress each outcome on indicator variables for assignments to the motivation treatment group, the monitoring treatment group, and the reminders treatment group, stratification dummies, and baseline controls selected by LASSO. Table D.1 shows the list of potential lasso controls. The adoption index (column 4) is standardized to have mean zero and standard deviation one in the control group. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Mechanisms. Recall that the reminder calls did not explicitly include any new information. They simply repeated information that principals had already received by email and via one previous phone call (which 95% of principals in the control group answered). Why then do the reminders increase implementation? We can think of two plausible mechanisms. First, reminders may simply bring to mind—at the right time—tasks which otherwise are forgotten, given the many responsibilities of school principals. Second, reminders might signal to the principals how important the task is to the Ministry, thereby increasing their motivation to carry out the task. In the language of the framework, these are the two terms of Equation (6): the attention term $G(v)\Delta a$ and the signaling term $a\, g(v)\Delta b^P$.

While the 'signaling importance' mechanism may well play some role, we suspect that the simple attentional / limited memory mechanism is the primary one in our setting. This parallels Lichand et al. (2026), who find that text messages merely redirecting parents' attention improve student outcomes as much as messages containing new information. In the first experiment, the incentives and evidence treatments plausibly also signaled the ministry's enthusiasm for the program, and yet had no effect. In the second experiment, reminders had a much larger effect (24 pp vs. 4.5 pp, $p < 0.001$) than a monitoring treatment that urged principals to implement the anti-bullying program and explicitly told them they were being monitored by the Ministry. Because the signaling channel operates through the same term as these direct shifters of b^S and b^P, their measured effects are informative about its plausible contribution: even the most explicit monitoring message moved implementation by only 4.5 percentage points. However, it is possible that receiving repeated calls signals importance and monitoring even more strongly than an email with stern monitoring language. While we view attention as the most plausible primary channel, we cannot rule out that signaling of the Ministry's priorities also contributes to the reminder effect.

The framework also disciplines magnitudes. If the pilot's workshops set attention $a \approx 1$, the pilot implementation rate identifies $\mathbb{E}[G(v)] \approx 0.83$; if the distribution of net benefits was broadly stable, the control group adoption rate of 37% implies baseline attention of roughly 0.45, and the reminder group's adoption rate of 58% implies roughly an attention parameter of 0.70. These numbers are approximate because the pilot rate is based on public schools in the original trial, whereas our sample includes only schools whose principals we reached at baseline. Read this way, standard communications left principals attending to the mandate less than half the time, and two additional calls closed nearly half of the remaining attention gap to the intensive pilot protocol—with no change in beliefs or preferences required.

Further evidence, albeit suggestive, comes from the exploratory heterogeneity analysis in Table A.14. Consistent with our framework and the view that reminders operate by alleviating attention constraints, we find larger treatment effects of reminders among principals with more managerial overload, using the variables defined

in Table 2.[20] Yet we find limited support for a related prediction that reminders should be more effective for principals who perceive greater net benefits from implementation. Contrary to this prediction, we do not find statistically significant interactions between the reminder treatment and our proxies for costs and benefits (being treated in the pilot experiment and the benefits index, respectively).

A related question is why two additional reminders were so effective when the control group had already received three emails and one phone call. Specifically, did the reminders work because they increased the frequency of contact or because they occurred closer to the AVE week, when principals could act more readily on the information? Our framework allows for both, and our design cannot separate these channels. In an exploratory analysis, we exploit variation in the timing of the *first* call, which all schools received in randomized order. Figure A.4 shows no relationship between the timing of the first call and program implementation among schools not assigned to reminders.[21] This provides (weak) evidence against the importance of timing. However, the first calls were all 13 to 7 days before AVE week, whereas reminder calls were during AVE week or in the preceding week. We cannot rule out that contacts much closer to the action date are much more effective than those a week or two earlier.

Downstream outcomes. In an exploratory analysis, we also estimate the effect of program adoption on the two outcomes of the original trial, dropout and test scores. Appendix F describes the data and the estimates. We link students to national enrollment and exam records and instrument photo submission with assignment to the reminders treatment.[22] The estimates go in the expected direction but—as expected—are imprecise: program adoption lowers dropout by 0.5 percent-

[20]Table A.16 reports correlations between the managerial overload variable and baseline school and principal characteristics, while Table A.17 shows that the heterogeneity results survive controlling for interactions between the treatment indicators and these baseline covariates.

[21]One caveat is that schools first contacted later were also less likely to receive a successfully completed call because we had to switch to the reminder calls on September 18 (second plot in Figure A.4).

[22]Our study was not designed to be powered for these downstream effects. Reminders, our only source of identifying variation in adoption, raise photo submission by 21 percentage points in the sample for which we observe these downstream outcomes. With this first stage, the minimum detectable effects of adoption at 80% power and a 5% significance level are 4.4 percentage points on dropout and 0.18 standard deviations on test scores.

age points (s.e. 1.6) and raises test scores by 0.03 standard deviations (s.e. 0.06). Our 95% confidence intervals contain both zero and the intent-to-treat estimates of the original trial, a 3 percentage point fall in dropout and a 0.05 standard deviation gain in test scores.

Cost-effectiveness. Section 2 identifies the cost per marginal implementing school, $\Delta C/\Delta q$, as the relevant statistic for comparing outreach technologies. We compute it for the reminders treatment, the only treatment that increased program adoption. The two additional follow-up calls to the 683 schools in the reminders treatment cost \$1,863.[23] The reminders treatment increased photo submission by 20.3 percentage points (Table 3), or about 139 additional schools implementing the program. Thus, each additional implementing school cost about \$13. By comparison, the in-person training used in the original trial cost \$57 per implementing school in 2015 and \$48 in 2016 (Berry et al., 2018), about four times as much.

We also compute the implied benefits using the estimates from the original trial, in which implementing AVE reduced dropout by 3 percentage points (Berry et al., 2018). Combined with the effect on adoption, this implies a 0.61-percentage-point reduction in dropout. With an average of 212 students per school, the reminders treatment reached about 145,000 students. Thus, the implied number of prevented dropouts is about 880, at a cost of about \$2 each. Barahona et al. (2017) find that preventing a dropout in the Dominican Republic raises the present value of formal earnings over the first seven years by \$734 to \$940. The implied benefit-cost ratio is between 350 and 440.

These calculations are illustrative. They combine our estimate of the effect on adoption with estimates of the effects on dropout and earnings from other samples. They understate costs, since they exclude both the fixed cost of producing the videos and the time school staff spend implementing the program. They also understate benefits, since they include only formal earnings and only the first seven years.

[23]The calls required 66 surveyor-days and 7 supervisor-days, a direct labor cost of \$1,579. The total includes the 18% overhead that the project budget adds to direct costs.

7 Discussion and Conclusion

In some ways, our study context is an ideal scenario for evidence-based policy. A government ministry commissioned a large-scale RCT to evaluate an intervention which was itself based on previous research (Jensen, 2010). The intervention proved effective and the ministry adopted the policy nationwide (Berry et al., 2018). But scaling up an effective policy requires more than selecting the right intervention—it requires ensuring that frontline bureaucrats actually implement it. Our study shows that even after the Ministry of Education mandated a proven educational program, only 37% of control group schools verifiably implemented the program, a stark contrast to the 83% compliance achieved through intensive in-person engagement in the original trial. Neither sharing research evidence, offering financial incentives, nor providing implementation assistance increased adoption. In contrast, simple reminder calls produced large gains in implementation, a finding replicated in a second experiment with a different program. These results suggest that limited attention among frontline bureaucrats is a central barrier to scaling effective policies.

Our findings carry practical implications. The intensive in-person workshops used in the original trial achieved near-universal implementation but are costly and difficult to scale. A far cheaper alternative—additional phone calls that simply remind principals to act—can recover a meaningful share of the implementation gap at a small fraction of the cost. Reminders are not a substitute for well-designed programs or good policy selection, but may be a low-cost tool to boost program implementation, especially when agents appear to perceive positive net benefits (for themselves) but still often do not follow through on implementation.

More broadly, our findings suggest that in settings where frontline bureaucrats manage many competing mandates, the traditional toolkit for improving performance—incentives, assistance, monitoring, and information provision—may be less effective than commonly assumed. Designing implementation protocols that account for the attentional demands facing these actors could be a useful complement to the growing emphasis on evidence-based policy selection.

Several limitations warrant discussion. First, further research in such settings is required to understand for which kinds of tasks reminders for frontline bureaucrats

would be helpful, whether reminders have persistent effects, and when they should be phased out. It seems plausible that reminders are helpful for new tasks which require one-time action or delegation. For tasks that require sustained effort or repeated actions, in contrast, reminders might not be as effective. Second, if attention is limited, reminders may divert principals from other valuable tasks. Altmann et al. (2022) show in a laboratory experiment that interventions directing attention to one decision can worsen choices in others, especially among those with fewer cognitive resources—plausibly the principals for whom our reminders were most effective. The welfare case for reminders thus rests on the reminded program being high-return. Note, however, that the pilot study found positive effects on students despite demanding substantially more of school principals' time and attention in the training workshops. Third, whether attentional barriers bind as strongly in other bureaucratic settings remains an open question.

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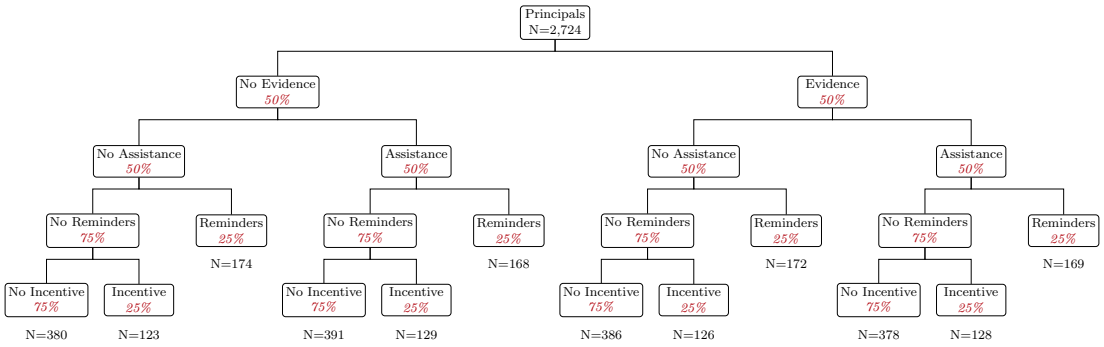
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Online Appendix

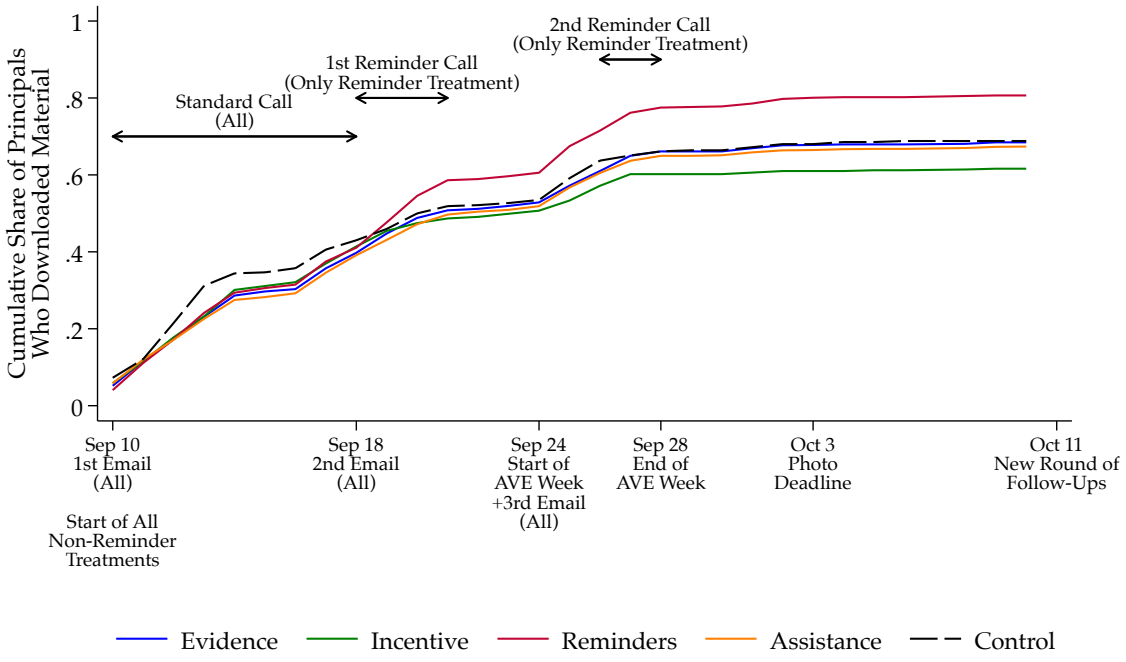
A Implementation Details

Figure A.1: Experimental Design



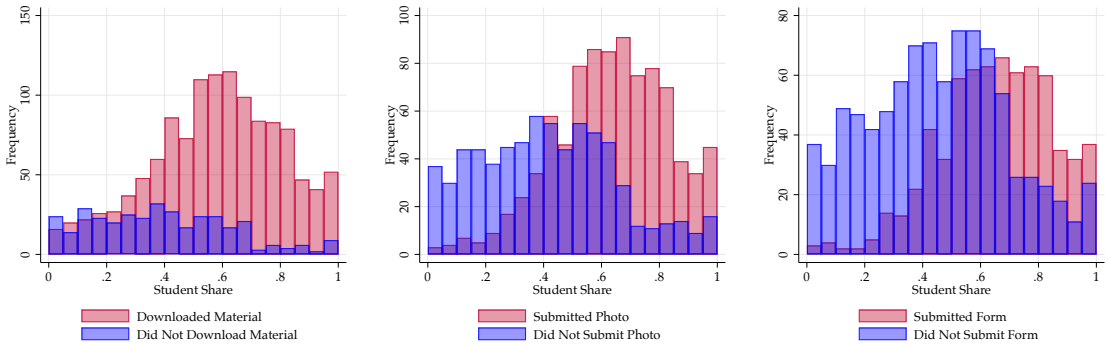
Notes: This figure shows the experimental design. We cross-randomized schools into treatment cells based on interactions between the "Evidence", "Assistance", "Reminders", and "Incentive" treatments. The Incentive × Reminders interaction was excluded from the design due to budget constraints.

Figure A.2: Download Rate Over Time by Treatment



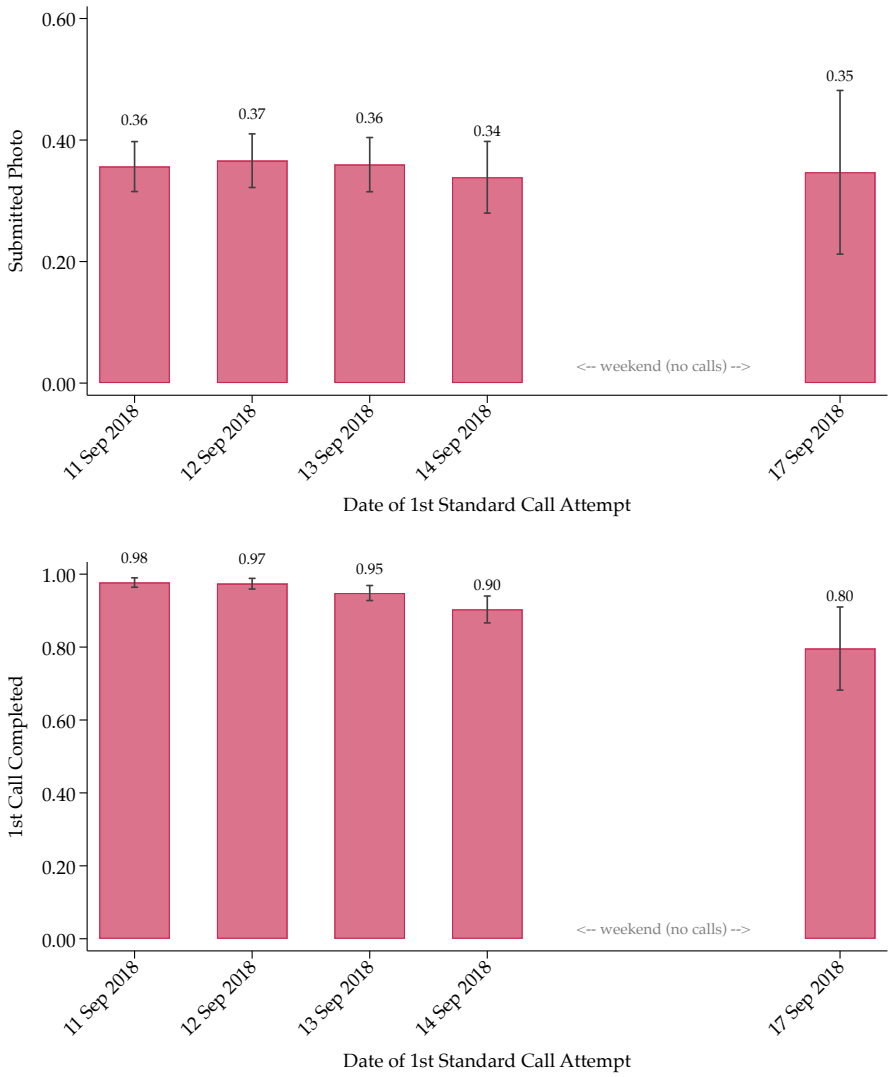
Notes: This figure shows the cumulative share of principals who downloaded the AVE campaign material over time. The blue line corresponds to the evidence treatment, the green line to the incentive treatment, the red line to the reminder treatment, the orange line to the assistance treatment, and the black dashed line to the control group. The kinks between September 15 and 16 and between September 22 and 23 correspond to weekends.

Figure A.3: The Distribution of Student Reports Across Different Adoption Measures



Notes: This figure shows the distribution across schools of the share of students reporting participation in the AVE campaign, including observing other students at their school participate. Distributions are shown separately according to whether each school met the corresponding administrative measure of implementation. The left panel splits schools by whether the principal downloaded the campaign material, the middle panel by whether the principal submitted a photo, and the right panel by whether the principal submitted the implementation form. In each panel, the red bars correspond to schools that met the administrative measure and the blue bars to schools that did not. Student-reported participation is based on questions included in the 2019 national examinations administered to 8th- and 12th-grade students. In all three panels, the null hypothesis that the two distributions are equal is rejected at $p < 0.001$.

Figure A.4: Adoption and Call Completion Rate by First Attempt Date



Notes: The first figure shows the share of schools that submitted a photo by the date of the first attempt to reach the school for the standard call offered to all principals. The second figure shows the share of schools whose principals were successfully reached with the standard call, including through follow-up attempts, by the date of the first attempt. The sample is restricted to schools assigned to the non-reminder treatment arm. We exclude schools first attempted on September 10 because, on that day, we prioritized schools with missing information. Whiskers indicate 95% confidence intervals.

Table A.1: Sample Selection

Sample (1)	Description (2)
4,351	Has less than five students in grades 7-12 in 2017 ($n = 162$)
4,189	Did not participate in at least one standardized exam between 2016 and 2018 ($n = 301$)
3,888	No contact details ($n = 7$)
3,881	Randomly select one school to drop if principal supervises more than one ($n = 119$)
3,762	Not reached at baseline ($n = 626$)
3,136	Principal has no access to email ($n = 343$)
2,793	Randomly select one school to drop if principals share the same email address ($n = 16$)
2,777	Did not meet criteria 1-3 in the updated administrative data ($n = 9$)
2,768	Part of pilot survey ($n = 44$)
2,724	Final sample

Notes: This table shows the sample selection criteria. Column (1) shows the remaining number of schools and column (2) describes the selection criteria. Out of 4,351 secondary schools in the Dominican Republic in total, 2,724 schools are included in our final sample.

Table A.2: Sample Comparison

	Universe of Secondary Schools (1)	Attempted at Baseline (2)	Reached at Baseline (3)	Final Sample (4)	p-value: (1) = (4) (5)
Has Internet Connection	0.685	0.692	0.702	0.715	0.007
Has Projector	0.720	0.746	0.755	0.767	0.000
Has Computer Lab	0.532	0.546	0.552	0.562	0.016
Has Electricity	0.978	0.981	0.982	0.987	0.004
Students in 7-12th Grade, 2018	183.385	197.097	204.224	211.906	0.000
Total PN Score 12th Grade, 2018	73.241	73.352	73.296	73.290	0.849
Total PN Score 8th Grade, 2016	69.072	69.015	68.987	68.988	0.700
Offers Extended School Day	0.438	0.458	0.463	0.465	0.024
Located in Urban Area	0.684	0.684	0.696	0.709	0.037
School Is Public	0.735	0.756	0.761	0.766	0.004
Treatment Group in AVE Pilot	0.282	0.314	0.320	0.324	0.000
Principal Has Multiple Positions	0.682	0.676	0.674	0.663	0.163
Principal Has No Secretary	0.280	0.256	0.252	0.243	0.011
Observations	4,351	3,762	3,136	2,724	

Notes: This table compares the characteristics of our final analysis sample to the broader universe of secondary schools in the Dominican Republic, following the sample selection steps described in Table A.1. Column (1) reports means for the universe of 4,351 secondary schools with students enrolled in grades 7–12 in 2018. Column (2) restricts to the 3,762 schools we attempted to reach for the baseline survey, column (3) to the 3,136 schools whose principals we reached at baseline, and column (4) to the 2,724 schools in our final sample. Each cell reports the mean of the characteristic among schools in the corresponding sample. The reported p-value tests equality of means between the samples in columns (1) and (4) using a stacked regression with robust standard errors.

Table A.3: Balance Check

	Control		Evidence		Incentive		Reminders		Assistance		
	Mean	St. D.	Coeff.	St. E.	Coeff.	St. E.	Coeff.	St. E.	Coeff.	St. E.	N
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Panel A: School Characteristics											
Has Internet Connection	0.732	[0.444]	-0.010	(0.017)	-0.014	(0.021)	-0.024	(0.019)	-0.018	(0.016)	2,717
Has Projector	0.758	[0.429]	-0.003	(0.017)	0.003	(0.021)	0.006	(0.019)	0.010	(0.016)	2,719
Has Computer Lab	0.567	[0.496]	-0.004	(0.019)	-0.007	(0.024)	0.020	(0.022)	-0.007	(0.018)	2,717
Has Electricity	0.989	[0.102]	0.000	(0.005)	0.000	(0.006)	0.004	(0.005)	0.000	(0.004)	2,719
Students in 7-12th Grade, 2018	233.029	[242.090]	-14.470	(8.948)	-0.047	(11.652)	-0.725	(9.660)	-7.388	(8.345)	2,724
Total PN Score 12th Grade, 2018	72.580	[7.211]	0.044	(0.424)	-0.599	(0.449)	-0.859*	(0.470)	0.090	(0.378)	1,324
Total PN Score 8th Grade, 2016	68.491	[7.046]	0.147	(0.331)	-0.491	(0.392)	-0.107	(0.369)	0.074	(0.301)	1,925
Shares Building with Another Sample School	0.129	[0.336]	-0.002	(0.012)	0.016	(0.016)	0.001	(0.013)	-0.018	(0.011)	2,724
Offers Extended School Day	0.497	[0.501]	-0.022	(0.018)	0.014	(0.022)	-0.004	(0.020)	-0.009	(0.017)	2,724
Located in Urban Area	0.742	[0.438]	-0.020	(0.017)	0.009	(0.022)	-0.026	(0.019)	0.013	(0.016)	2,358
School Is Public	0.800	[0.401]	-0.001	(0.016)	0.010	(0.020)	0.018	(0.018)	-0.023	(0.015)	2,724
Treatment Group in AVE Pilot	0.313	[0.464]	0.006	(0.019)	0.014	(0.024)	0.011	(0.021)	-0.013	(0.018)	2,724
Panel B: Principal Characteristics											
Experience of Principal	20.139	[7.333]	-0.951***	(0.356)	1.000**	(0.458)	0.438	(0.409)	-0.200	(0.336)	1,993
Principal Has Post-Graduate Degree	0.567	[0.496]	-0.010	(0.023)	-0.009	(0.030)	-0.045*	(0.026)	-0.008	(0.022)	1,955
Principal Supervises Multiple Schools	0.093	[0.290]	0.012	(0.012)	-0.011	(0.014)	0.009	(0.014)	-0.013	(0.011)	2,710
Principal Checks Emails Daily	0.597	[0.491]	0.006	(0.020)	-0.039	(0.025)	-0.049**	(0.023)	0.001	(0.019)	2,630
Believes Penalty after Non-Compliance is Very Likely	0.537	[0.499]	-0.006	(0.020)	0.034	(0.025)	0.005	(0.023)	0.014	(0.019)	2,723
Principal's Main Motivation is Professional Success	0.455	[0.499]	-0.003	(0.020)	-0.035	(0.025)	0.023	(0.023)	-0.001	(0.019)	2,723
Believes Extra Info Would Have An Effect	0.826	[0.379]	-0.015	(0.014)	-0.018	(0.019)	0.018	(0.015)	0.006	(0.013)	2,724
Believes Students Undervalue Education	0.551	[0.498]	-0.002	(0.021)	0.039	(0.026)	0.030	(0.024)	-0.008	(0.020)	2,630
Believes Lack of Assistance is Major Barrier	0.392	[0.489]	0.014	(0.020)	-0.028	(0.025)	0.012	(0.023)	-0.019	(0.019)	2,724
Number of Valid Emails	1.387	[0.634]	0.029	(0.027)	-0.038	(0.033)	-0.013	(0.031)	-0.011	(0.025)	2,724
Principal Has Multiple Positions	0.656	[0.476]	-0.017	(0.022)	0.038	(0.028)	0.042*	(0.025)	0.026	(0.021)	1,994
Principal Has No Secretary	0.264	[0.442]	0.030	(0.024)	-0.067**	(0.028)	-0.021	(0.026)	0.010	(0.022)	1,499
Joint Test p-value			0.59		0.18		0.36		0.86		

Notes: This table shows balance for baseline covariates. Panel A reports on school-level outcomes and Panel B on principal-level outcomes. Differences in sample sizes across variables reflect missing data. Column (1) reports the control group mean of the dependent variable. Columns (3), (5), (7) and (9) report the difference in the dependent variable from OLS regressions of each outcome on indicator variables for assignments to the evidence treatment group, the incentive treatment group, the reminders treatment group and the assistance treatment group, and stratification dummies. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.4: Intermediate Implementation Outcomes

	1st Email		2nd Email		3rd Email				
	Sent	Opened	Sent	Opened	Sent	Opened	Standard Call	1st Reminder Call	2nd Reminder Call
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Evidence	-0.000	0.030*	-0.008*	0.032*	-0.001	0.027	-0.017*	0.004	-0.011*
	(0.006)	(0.016)	(0.005)	(0.018)	(0.006)	(0.018)	(0.009)	(0.005)	(0.007)
Incentive	-0.003	-0.019	-0.004	0.007	-0.004	-0.041*	-0.008	0.001	0.002
	(0.007)	(0.021)	(0.006)	(0.023)	(0.007)	(0.023)	(0.012)	(0.001)	(0.002)
Reminders	-0.013*	0.007	-0.009	0.066***	-0.002	0.010	0.004	0.942***	0.877***
	(0.007)	(0.018)	(0.006)	(0.019)	(0.006)	(0.020)	(0.010)	(0.009)	(0.012)
Assistance	0.001	-0.023	0.004	-0.021	0.002	-0.044**	0.003	0.002	0.002
	(0.006)	(0.015)	(0.004)	(0.017)	(0.005)	(0.017)	(0.009)	(0.004)	(0.006)
Sample	Full	Full	Full	Full	Full	Full	Full	Full	Full
Mean of Control Group	0.982	0.750	0.995	0.687	0.982	0.692	0.955	0.000	0.000
Observations	2724	2724	2724	2724	2724	2724	2724	2724	2724

Notes: This table shows differences in implementation outcomes by treatment arm. For phone calls (standard or reminders), we report probabilities of successfully reaching the schools. We regress each outcome on indicator variables for assignments to the evidence treatment group, the incentive treatment group, the reminders treatment group and the assistance treatment group, stratification dummies, and baseline controls selected by LASSO. Table A.3 shows the list of potential lasso controls. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.5: Multiple Hypothesis Correction

	Index Components			
	Primary Outcome	Secondary Outcomes		
	Submitted Photo	Downloaded Material	Submitted Form	Adoption Index (in Std)
	(1)	(2)	(3)	(4)
Reminders	0.001	0.001	0.001	0.001
Evidence	0.981	0.464	0.069	0.220
Incentive	0.981	0.281	0.024	0.378
Assistance	0.981	0.059	0.134	0.157

Notes: This table reports sharpened false discovery rate (FDR) q-values corresponding to the estimates in Table 3. Q-values are computed following Benjamini et al. (2006) and control the expected proportion of false discoveries among all rejected hypotheses. For each outcome column, the correction is applied across the family of four hypotheses corresponding to the four treatment arms.

Table A.6: Pre-Analysis Plan and Implemented Analysis

Table	Specified in PAP	Deviations
Table 3 - AVE Campaign Adoption		
(1) Submitted Photo	*"We will measure whether the principal uploads evidence of completion as required in the program instructions."*	None.
(2) Downloaded Material	*"The secondary outcomes - which are better thought of as intermediate steps rather than separate outcomes - are whether the principal clicked on the landing page in the email, and whether they downloaded the videos."*	We do not separately report results on whether the principals clicked on the landing page, as this closely mirrors the download rate.
(3) Submitted Form		Not explicitly mentioned in the first PAP, as we were still working out the reporting process at the time of submission
(4) Adoption Index		Not pre-specified.
Table 4 - Anti-Bullying Campaign Adoption		
(1) Submitted Photo	*'Principals were required to send pictures of students performing the task to a phone number of the Ministry."*	None.
(2) Clicked on Declaration	*"The secondary outcomes - which are better thought of as intermediate steps rather than separate outcomes - are whether the principal clicked on the email attachment containing the instructions of the task."*	None.
(3) Submitted Form	*"We will measure whether the principal uploads evidence of completion as required in the program instructions."*	The PAP lists it as a primary outcome but we use it as a secondary outcome to be consistent with Table 3.
(4) Adoption Index		Not pre-specified.
Additional Analysis		
Cutoff date	An extension to the first PAP reports the treatment effects as of October 11 and notes that we will report additional results separately after extending the deadline and conducting another round of outreach to principals.	None.
Student reports	The first PAP explicitly lists under primary outcomes that *'we will attempt to acquire independent validation of this through student surveys, although the feasibility of doing so is not clear at the time of this registration"*.	While we list student reports as primary outcomes in the PAP, we report these results separately in Table A.11 because the outcomes were collected with a substantial delay and only after additional follow-ups with principals.
Multiple hypothesis test corrections		Not pre-specified.
Spillover effects	*"High evidence share vs. low evidence share."*	We do not examine spillovers since we find no direct effects of the evidence treatment.

Notes: The project was pre-registered under AEA registry ID AEARCTR-0003324 and AEARCTR-0003621.

Table A.7: Effect of AVE Pilot Assignment on AVE Campaign Adoption

	Submitted Photo (1)	Downloaded Material (2)	Submitted Form (3)	Adoption Index (4)
Treatment Group in AVE Pilot	-0.027	0.010	-0.011	-0.016
	(0.028)	(0.027)	(0.026)	(0.056)
Outcome Mean	0.437	0.647	0.304	0.079
Observations	1351	1351	1351	1351

Notes: This table shows the effect of having been assigned to the treatment group in the AVE pilot on AVE campaign adoption during the scale-up. We regress each outcome on an indicator for assignment to the treatment group in the AVE pilot and stratification dummies. All regressions are estimated by OLS and robust standard errors are reported in parentheses.

Table A.8: Treatment Effects on AVE Campaign Adoption With Alternative Sample Restrictions

	Index Components			
	Primary Outcome	Secondary Outcomes		
	Submitted Photo	Downloaded Material	Submitted Form	Adoption Index (in Std)
	(1)	(2)	(3)	(4)
Panel A: Without Pilot Schools				
Reminders	0.187***	0.130***	0.095***	0.342***
	(0.027)	(0.024)	(0.025)	(0.053)
Evidence	-0.020	-0.019	-0.032	-0.063
	(0.023)	(0.022)	(0.022)	(0.048)
Incentive	0.016	-0.040	0.041	-0.000
	(0.029)	(0.029)	(0.028)	(0.063)
Assistance	-0.007	-0.033	-0.009	-0.049
	(0.022)	(0.021)	(0.020)	(0.045)
Mean of Control Group	0.387	0.697	0.295	0.042
Observations	1843	1843	1843	1843
Panel B: Without Private Schools				
Reminders	0.199***	0.149***	0.127***	0.412***
	(0.025)	(0.021)	(0.024)	(0.048)
Evidence	-0.004	0.008	-0.018	-0.009
	(0.022)	(0.020)	(0.021)	(0.045)
Incentive	0.004	-0.008	0.055**	0.055
	(0.027)	(0.027)	(0.026)	(0.057)
Assistance	-0.010	-0.030	-0.018	-0.057
	(0.021)	(0.019)	(0.019)	(0.041)
Mean of Control Group	0.395	0.691	0.263	0.011
Observations	2086	2086	2086	2086

Notes: This table replicates Table 3 with alternative sample restrictions. Panel A excludes schools that were assigned to the treatment group in the original AVE pilot and Panel B excludes private schools. We regress each outcome on indicator variables for assignments to the evidence treatment group, the incentive treatment group, the reminders treatment group and the assistance treatment group, stratification dummies, and baseline controls selected by LASSO. Table A.3 shows the list of potential lasso controls. The adoption index (column 4) is standardized to have mean zero and standard deviation one in the control group. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.9: Effects of All Treatment Cells on AVE Campaign Adoption

	Index Components			
	Primary Outcome	Secondary Outcomes		
	Submitted Photo	Downloaded Material	Submitted Form	Adoption Index (in Std)
	(1)	(2)	(3)	(4)
Evidence	0.002	-0.014	-0.033	-0.029
	(0.034)	(0.033)	(0.032)	(0.070)
Incentive	-0.032	-0.041	0.051	-0.011
	(0.047)	(0.048)	(0.047)	(0.104)
Reminders	0.196***	0.162***	0.146***	0.446***
	(0.044)	(0.037)	(0.043)	(0.086)
Assistance	0.004	-0.011	0.001	0.003
	(0.034)	(0.032)	(0.031)	(0.070)
Evidence + Incentive	0.029	-0.037	0.076	0.059
	(0.048)	(0.048)	(0.047)	(0.105)
Evidence + Reminders	0.198***	0.141***	0.104**	0.375***
	(0.045)	(0.037)	(0.042)	(0.086)
Evidence + Assistance	-0.044	-0.074**	-0.047	-0.142**
	(0.034)	(0.034)	(0.031)	(0.071)
Incentive + Assistance	0.066	-0.092*	0.037	-0.007
	(0.050)	(0.049)	(0.046)	(0.108)
Reminders + Assistance	0.158***	0.062	0.094**	0.264***
	(0.044)	(0.039)	(0.042)	(0.086)
Evidence + Incentive + Assistance	-0.058	-0.022	-0.021	-0.081
	(0.047)	(0.047)	(0.043)	(0.098)
Evidence + Reminders + Assistance	0.221***	0.136***	0.102**	0.390***
	(0.045)	(0.038)	(0.043)	(0.087)
Mean of Control Group	0.374	0.687	0.271	0.000
Observations	2724	2724	2724	2724

Notes: This table shows the treatment effects on different adoption measures of the AVE campaign. We regress each outcome on indicator variables for assignments to each treatment cell, stratification dummies, and baseline controls selected by LASSO. Table A.3 shows the list of potential lasso controls. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.10: Pairwise Comparisons Across Treatment Cells

	Estimate/ Difference (1)	*p*-val (2)	Ex-post MDE (3)		Estimate/ Difference (4)	*p*-val (5)	Ex-post MDE (6)
Panel A: Pooled Treatment Effects (Table 3)							
Evidence	-0.011	0.553	0.054	Reminders	0.203	0.000***	0.061
Incentive	0.010	0.660	0.067	Assistance	-0.012	0.514	0.050
Panel B: Cross-treatment Comparisons (Table 3)							
Evidence vs. Incentive	-0.022	0.477	0.086	Incentive vs. Reminders	-0.192	0.000***	0.077
Evidence vs. Reminders	-0.214	0.000***	0.081	Incentive vs. Assistance	0.022	0.454	0.083
Evidence vs. Assistance	0.000	0.989	0.074	Reminders vs. Assistance	0.214	0.000***	0.079
Panel C: Treatment Cells vs. Control (Table A.9)							
Evidence Only	0.002	0.958	0.096	Ev+Ass	-0.044	0.199	0.096
Incentive Only	-0.032	0.489	0.131	Inc+Ass	0.066	0.188	0.141
Reminders Only	0.196	0.000***	0.123	Rem+Ass	0.158	0.000***	0.124
Assistance Only	0.004	0.905	0.095	Ev+Inc+Ass	-0.058	0.215	0.131
Ev+Inc	0.029	0.545	0.133	Ev+Rem+Ass	0.221	0.000***	0.126
Ev+Rem	0.198	0.000***	0.126				
Panel D: Pairwise Cross-cut Comparisons (Table A.9)							
Evidence Only vs. Incentive Only	0.034	0.465	0.131	Assistance Only vs. Ev+Rem	-0.194	0.000***	0.125
Evidence Only vs. Reminders Only	-0.195	0.000***	0.123	Assistance Only vs. Ev+Ass	0.048	0.156	0.095
Evidence Only vs. Assistance Only	-0.002	0.947	0.095	Assistance Only vs. Inc+Ass	-0.062	0.216	0.141
Evidence Only vs. Ev+Inc	-0.027	0.560	0.130	Assistance Only vs. Rem+Ass	-0.154	0.001***	0.125
Evidence Only vs. Ev+Rem	-0.196	0.000***	0.124	Assistance Only vs. Ev+Inc+Ass	0.062	0.183	0.130
Evidence Only vs. Ev+Ass	0.046	0.164	0.092	Assistance Only vs. Ev+Rem+Ass	-0.217	0.000***	0.125
Evidence Only vs. Inc+Ass	-0.064	0.202	0.141	Ev+Inc vs. Ev+Rem	-0.169	0.002***	0.155
Evidence Only vs. Rem+Ass	-0.156	0.000***	0.125	Ev+Inc vs. Ev+Ass	0.073	0.116	0.130
Evidence Only vs. Ev+Inc+Ass	0.060	0.194	0.129	Ev+Inc vs. Inc+Ass	-0.037	0.534	0.169
Evidence Only vs. Ev+Rem+Ass	-0.219	0.000***	0.122	Ev+Inc vs. Rem+Ass	-0.129	0.020**	0.156
Incentive Only vs. Reminders Only	-0.229	0.000***	0.153	Ev+Inc vs. Ev+Inc+Ass	0.087	0.125	0.159
Incentive Only vs. Assistance Only	-0.037	0.437	0.132	Ev+Inc vs. Ev+Rem+Ass	-0.192	0.000***	0.151
Incentive Only vs. Ev+Inc	-0.061	0.284	0.160	Ev+Rem vs. Ev+Ass	0.242	0.000***	0.124
Incentive Only vs. Ev+Rem	-0.231	0.000***	0.155	Ev+Rem vs. Inc+Ass	0.132	0.023**	0.163
Incentive Only vs. Ev+Ass	0.012	0.804	0.132	Ev+Rem vs. Rem+Ass	0.040	0.447	0.149
Incentive Only vs. Inc+Ass	-0.099	0.099*	0.168	Ev+Rem vs. Ev+Inc+Ass	0.256	0.000***	0.152
Incentive Only vs. Rem+Ass	-0.190	0.001***	0.153	Ev+Rem vs. Ev+Rem+Ass	-0.023	0.665	0.148
Incentive Only vs. Ev+Inc+Ass	0.025	0.655	0.159	Ev+Ass vs. Inc+Ass	-0.110	0.029**	0.141
Incentive Only vs. Ev+Rem+Ass	-0.254	0.000***	0.154	Ev+Ass vs. Rem+Ass	-0.202	0.000***	0.125
Reminders Only vs. Assistance Only	0.192	0.000***	0.122	Ev+Ass vs. Ev+Inc+Ass	0.014	0.765	0.129
Reminders Only vs. Ev+Inc	0.168	0.002***	0.153	Ev+Ass vs. Ev+Rem+Ass	-0.265	0.000***	0.122
Reminders Only vs. Ev+Rem	-0.002	0.973	0.147	Inc+Ass vs. Rem+Ass	-0.091	0.116	0.163
Reminders Only vs. Ev+Ass	0.240	0.000***	0.123	Inc+Ass vs. Ev+Inc+Ass	0.124	0.037**	0.167
Reminders Only vs. Inc+Ass	0.130	0.023**	0.160	Inc+Ass vs. Ev+Rem+Ass	-0.155	0.008***	0.163
Reminders Only vs. Rem+Ass	0.039	0.462	0.147	Rem+Ass vs. Ev+Inc+Ass	0.216	0.000***	0.154
Reminders Only vs. Ev+Inc+Ass	0.254	0.000***	0.152	Rem+Ass vs. Ev+Rem+Ass	-0.063	0.234	0.149
Reminders Only vs. Ev+Rem+Ass	-0.025	0.638	0.147	Ev+Inc+Ass vs. Ev+Rem+Ass	-0.279	0.000***	0.152
Assistance Only vs. Ev+Inc	-0.025	0.600	0.132				

Notes: This table reports pairwise comparisons of AVE campaign adoption across all treatment cells based on the regression in column (1) in Tables 3 and A.9. The "Difference" column reports the difference in adoption rates between the two cells, and the "p-value" column the p-value from a test of their equality. The "Ex-post MDE" column reports the ex-post minimum detectable effect (80% power, 5% two-sided) for each comparison, in the same units as the "Difference" column. Treatment cells are abbreviated Ev (Evidence), Inc (Incentive), Rem (Reminders), and Ass (Assistance). Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.11: Effects on Student Recall of AVE Videos a Year Later

	Student Share (1)	Student Share $\geq$ 50% (2)
Reminders	0.034***	0.063**
	(0.013)	(0.027)
Evidence	0.020*	0.005
	(0.011)	(0.024)
Incentive	-0.035***	-0.077***
	(0.014)	(0.029)
Assistance	-0.018*	-0.019
	(0.010)	(0.022)
Mean of Control Group	0.538	0.617
Observations	1588	1588

Notes: This table reports the effects of the AVE campaign adoption based on student reports. In Column (1), the outcome is the share of students who report either participating in the AVE campaign themselves or observing participation by other students in their school. In Column (2), the outcome is an indicator equal to one if this share is at least 50%. We regress each outcome on indicator variables for assignments to the evidence treatment group, the incentive treatment group, the reminders treatment group and the assistance treatment group, stratification dummies, and baseline controls selected by LASSO. Table A.3 shows the list of potential lasso controls. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.12: Treatment Effects on AVE Campaign Adoption by December 1

	Index Components			
	Primary Outcome	Secondary Outcomes		
	Submitted Photo	Downloaded Material	Submitted Form	Adoption Index (in Std)
	(1)	(2)	(3)	(4)
Reminders	0.165***	0.100***	0.120***	0.309***
	(0.021)	(0.017)	(0.022)	(0.041)
Evidence	0.009	-0.018	-0.012	-0.027
	(0.019)	(0.017)	(0.019)	(0.039)
Incentive	-0.033	-0.043*	-0.004	-0.071
	(0.024)	(0.023)	(0.024)	(0.051)
Assistance	-0.027	-0.032**	-0.026	-0.073**
	(0.018)	(0.016)	(0.018)	(0.036)
Mean of Control Group	0.553	0.763	0.450	-0.000
Observations	2724	2724	2724	2724

Notes: This table shows the treatment effects on different adoption measures of the AVE campaign. We regress each outcome on indicator variables for assignments to the evidence treatment group, the incentive treatment group, the reminders treatment group and the assistance treatment group, stratification dummies, and baseline controls selected by LASSO. Table A.3 shows the list of potential lasso controls. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.13: Alternative Outcome Combinations

	Full Sample			Student Report Sample		
	Photo	Download & (Form or Photo)	Download & Form & Photo	Photo	Download & ((Form & Photo) or Student Share $\geq 50\%$)	Download & Form & Photo & Student Share $\geq 50\%$
	(1)	(2)	(3)	(4)	(5)	(6)
Reminders	0.203***	0.211***	0.113***	0.173***	0.150***	0.081***
	(0.022)	(0.022)	(0.020)	(0.029)	(0.028)	(0.024)
Evidence	-0.011	-0.024	-0.035**	-0.008	-0.025	-0.041**
	(0.019)	(0.019)	(0.017)	(0.025)	(0.026)	(0.021)
Incentive	0.010	0.006	0.065***	0.028	-0.000	0.057**
	(0.024)	(0.024)	(0.022)	(0.032)	(0.032)	(0.026)
Assistance	-0.012	-0.023	-0.021	-0.005	-0.050**	0.009
	(0.018)	(0.018)	(0.016)	(0.024)	(0.024)	(0.019)
Mean of Control Group	0.374	0.395	0.239	0.383	0.563	0.189
Observations	2724	2724	2724	1588	1588	1588

Notes: This table shows treatment effects on alternative definitions of AVE campaign adoption that combine our different implementation measures. In each column the dependent variable is an indicator equal to one if the school met the combination of measures given in the column header, where "Photo," "Download," and "Form" refer to the principal submitting a photo, downloading the material, and submitting the implementation form, and "Student Share $\geq 50\%$" indicates that at least half of surveyed students reported participation. Columns (1)–(3) use the full sample, while columns (4)–(6) restrict to the subsample of schools for which student reports are available. We regress each outcome on indicator variables for assignments to the evidence treatment group, the incentive treatment group, the reminders treatment group and the assistance treatment group, stratification dummies, and baseline controls selected by LASSO. Table A.3 shows the list of potential lasso controls. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.14: Heterogeneity Analysis

Heterogeneity Variable:	Primary Outcome: Submitted Photo		
	Managerial Overload Index	Benefits Index	Implementation Costs
	(1)	(2)	(3)
Evidence	-0.000	-0.015	-0.021
	(0.036)	(0.047)	(0.023)
Incentive	0.028	0.009	0.008
	(0.046)	(0.060)	(0.029)
Reminders	0.135***	0.208***	0.184***
	(0.042)	(0.058)	(0.027)
Assistance	-0.014	-0.022	-0.006
	(0.034)	(0.046)	(0.022)
Evidence × Heterogeneity Var.	-0.004	0.007	0.034
	(0.055)	(0.060)	(0.039)
Incentive × Heterogeneity Var.	-0.040	0.001	0.009
	(0.070)	(0.078)	(0.052)
Reminders × Heterogeneity Var.	0.123*	-0.007	0.060
	(0.067)	(0.075)	(0.048)
Assistance × Heterogeneity Var.	0.009	0.015	-0.020
	(0.054)	(0.060)	(0.039)
Heterogeneity Var.	-0.128**	0.027	-0.049
	(0.050)	(0.056)	(0.038)
Mean of Control Group	0.394	0.374	0.374
Mean of Heterogeneity Var.	0.503	0.708	0.323
Observations	2058	2724	2724

Notes: This table shows heterogeneous treatment effects on whether the principal submitted a photo of the implementation of the AVE campaign in their school. We regress the outcome on treatment indicators, the relevant index, their interactions, stratification dummies, and baseline controls selected by LASSO. Table A.3 shows the list of potential lasso controls. Table A.15 reports the results for each index component. Implementation costs is an indicator variable for whether the school was in the treatment group in the AVE pilot. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.15: Heterogeneity Analysis by Index Components

	Primary Outcome: Submitted Photo			
	Managerial Overload Index Components		Benefits Index Components	
Heterogeneity Variable:	Principal Has Multiple Positions	Principal Has No Secretary	Believes Info Would Have Effect	Believes Penalty is Very Likely
	(1)	(2)	(3)	(4)
Evidence	0.012	-0.003	-0.005	-0.014
	(0.038)	(0.030)	(0.050)	(0.028)
Incentive	-0.005	0.055	0.009	0.009
	(0.050)	(0.038)	(0.062)	(0.036)
Reminders	0.108**	0.214***	0.177***	0.213***
	(0.046)	(0.034)	(0.064)	(0.033)
Assistance	0.010	-0.053*	-0.031	-0.014
	(0.037)	(0.028)	(0.049)	(0.027)
Evidence × Heterogeneity Var.	-0.015	-0.018	-0.006	0.006
	(0.046)	(0.058)	(0.053)	(0.037)
Incentive × Heterogeneity Var.	0.029	-0.066	0.002	0.001
	(0.061)	(0.080)	(0.067)	(0.049)
Reminders × Heterogeneity Var.	0.149***	-0.029	0.029	-0.020
	(0.056)	(0.069)	(0.068)	(0.045)
Assistance × Heterogeneity Var.	-0.032	0.077	0.022	0.003
	(0.046)	(0.057)	(0.053)	(0.037)
Heterogeneity Var.	-0.065	-0.138***	-0.009	0.024
	(0.044)	(0.053)	(0.048)	(0.036)
Mean of Control Group	0.391	0.377	0.374	0.374
Mean of Heterogeneity Var.	0.663	0.243	0.862	0.553
Observations	1994	1499	2724	2723

Notes: This table shows heterogeneous treatment effects on whether the principal uploaded a photo of the implementation of the AVE campaign in their school for each index component. We regress the outcome on indicator variables for assignments to the evidence treatment group, the incentive treatment group, the reminders treatment group, the assistance treatment group, the heterogeneity dummy, interactions between the treatment group variables and the heterogeneity dummy, stratification dummies, and baseline controls selected by LASSO. Table A.3 shows the list of potential lasso controls. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.16: Correlations Between Managerial Overload and Other Variables

	Managerial Overload Index (1)
Has Internet Connection	-0.004
	(0.008)
Has Projector	-0.013
	(0.008)
Has Computer Lab	-0.011
	(0.009)
Has Electricity	0.001
	(0.007)
Students in 7-12th Grade, 2018	-0.042***
	(0.010)
Shares Building with Another Sample School	-0.008
	(0.008)
Offers Extended School Day	-0.114***
	(0.009)
Located in Urban Area	-0.045***
	(0.009)
Treatment Group in AVE Pilot	0.009
	(0.008)
Total PN Score 12th Grade, 2018	-0.017
	(0.015)
Total PN Score 8th Grade, 2016	0.014
	(0.012)
Experience of Principal	0.046***
	(0.009)
Principal Has Post-Graduate Degree	-0.011
	(0.008)
Principal Supervises Multiple Schools	0.036***
	(0.008)
Principal Checks Emails Daily	-0.014*
	(0.008)
Believes Lack of Assistance is Major Barrier	0.007
	(0.008)
Principal's Main Motivation is Professional Success	0.017**
	(0.008)
Believes Extra Info Would Have An Effect	0.014*
	(0.008)
Believes Students Undervalue Education	0.003
	(0.008)
Number of Valid Emails	-0.001
	(0.008)
Believes Penalty after Non-Compliance is Very Likely	0.012
	(0.008)
Observations	2058

Notes: This table describes how the managerial overload index relates to observable school and principal characteristics in a joint regression. The sample is restricted to public schools. The regression is estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

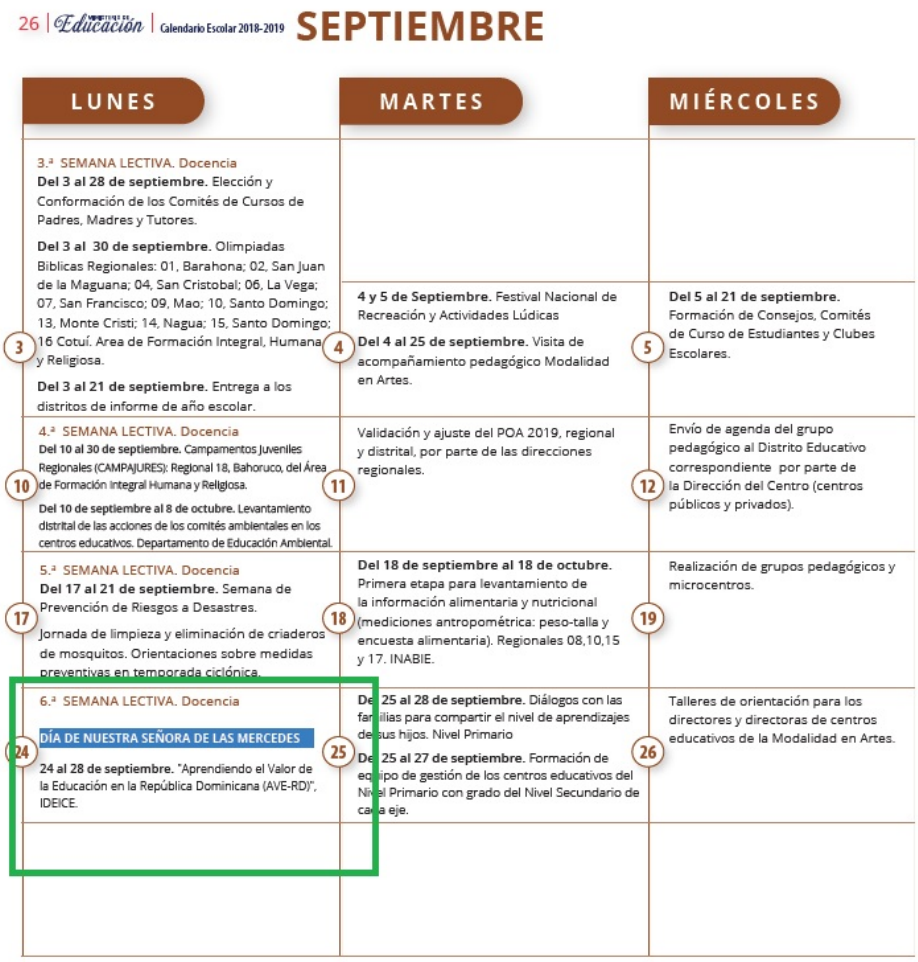
Table A.17: Robustness of Heterogeneity Analysis

Heterogeneity Variable:	Primary Outcome: Submitted Photo		
	Managerial Overload Index	Benefits Index	Implementation Costs
	(1)	(2)	(3)
Evidence	-0.011	-0.016	-0.022
	(0.060)	(0.049)	(0.025)
Incentive	-0.139**	0.024	0.031
	(0.069)	(0.061)	(0.032)
Reminders	0.028	0.232***	0.187***
	(0.071)	(0.062)	(0.029)
Assistance	0.015	-0.027	-0.008
	(0.058)	(0.049)	(0.024)
Evidence × Heterogeneity Var.	0.026	0.008	0.037
	(0.069)	(0.064)	(0.046)
Incentive × Heterogeneity Var.	0.064	-0.002	-0.030
	(0.085)	(0.080)	(0.060)
Reminders × Heterogeneity Var.	0.167**	-0.033	0.067
	(0.084)	(0.080)	(0.056)
Assistance × Heterogeneity Var.	0.041	0.013	-0.029
	(0.067)	(0.064)	(0.046)
Mean of Control Group	0.394	0.374	0.374
Mean of Heterogeneity Var.	0.503	0.708	0.323
Observations	2058	2724	2724

Notes: This table assesses the robustness of the heterogeneity analysis. The outcome is whether the principal uploaded a photo of the implementation of the AVE campaign in their school. Each regression follows Table A.14 but additionally includes the full set of baseline covariates reported in Table A.3 and their interactions with the treatment indicators. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

B Additional Material for AVE Campaign

Figure B.1: Official Academic Calendar



Notes: This figure shows the official academic calendar for the 2018-2019 school year in the Dominican Republic. The calendar states that the AVE campaign should be implemented between September 24 and 28.

Figure B.2: Intervention Brochure for Control Group



Notes: This figure shows the brochure that was attached to the first email for the control group.

Figure B.3: Intervention Brochure for Evidence Treatment Group



Notes: This figure shows the brochure that was attached to the first email for the evidence group.

C Email and Call Scripts for the AVE Campaign

Email Script (Control Group)

Greetings.

As you are aware, the week of September 25 to 28 marks the "Learning the Value of Education in the Dominican Republic" (AVE-RD) week, as outlined in the 2018-2019 MINERD school calendar.[24] AVE-RD is an informational campaign being promoted by MINERD through the Directorate of Guidance and Psychology, in collaboration with IDEICE.

The campaign consists of four videos in a telenovela format, with a total running time of one hour; these videos must be screened for students in grades 1 through 6 of secondary education at all educational centers throughout the country. The videos focus on informing students about the economic, social, and personal benefits of education in an engaging and relatable manner.

On the webpage XXX, you may download the videos and other materials that will guide you—along with your management team and Guidance and Psychology staff—through the steps required to fulfill this mandate.

Additionally, this email includes a set of instructions for downloading the videos and manuals, as well as an informational brochure regarding this campaign. Please download both documents by clicking on the download buttons labeled "Descargar Folleto" (Download Brochure) and "Descargar Instructivo" (Download Instructions), and review them carefully.

We appreciate your collaboration and are confident that you will successfully carry out this campaign. Please reply to this email to confirm receipt of this message.

Director of Guidance and Psychology, MINERD
Executive Director, IDEICE

[24] While the official academic calendar lists September 24–28 as the AVE week, the emails and phone calls refer to September 25–28, since September 24 was a religious holiday.

Call Protocols

The calls were placed by public employees working at an established Ministry of Education call center. A total of 33 callers conducted the standard calls, while 10 conducted the two reminder calls. Schools were contacted in randomized order during regular working hours, with up to three call attempts per school.

1st Call (Full Sample)

Everyone:

Hello, my name is XXX and I am calling on behalf of the Ministry of Education to give you an important message. Do you have a couple of minutes?

Are you currently in charge of the school XXX with Center Code XXX?

We are calling to remind you about the delivery of an email regarding the informational campaign Learning the Value of Education in the Dominican Republic (AVE-RD) on behalf of MINERD and IDEICE.

I first need to verify some of your information and information about your school. Are you XXX?

This email was sent by the Ministry of Education and IDEICE on behalf of Minerva Perez, Director of the Directorate of Guidance and Psychology, on Monday, September 10, and contains relevant information so that your school can implement the campaign.

I wanted to notify you that we sent the email to the following accounts: XXX

Have you been able to see this email?

As mentioned, this campaign consists of a series of four videos in telenovela format, which you must show to secondary school students in your school between September 25 and September 28, as established in the MINERD 2018-2019 school calendar.

These videos provide information about the economic, social, and personal benefits of finishing school.

Evidence Treatment Arm:

We would like to tell you that a study conducted in 2,500 schools in the Dominican Republic showed that there are proven benefits of implementing this campaign in your educational center.

Schools that showed these videos significantly improved their national exam scores by 0.6 points in total, and reduced their school dropout rates.

This study concludes that if all schools in the country showed these videos, the number of secondary school dropouts would decrease by 6,500 students per year.

Would you like me to repeat the information?

Incentive Treatment Arm:

If you implement the AVE-RD campaign in your school, you will have the opportunity to win a 7-inch Samsung tablet to help you manage your educational center. One out of every 10 people who participate in the raffle will win a tablet.

You will be able to participate in this raffle only if you implement the campaign in your school as established in the instructions in the email we sent you.

Would you like me to repeat the information?

Assistance Treatment Arm:

We would like to mention that a support center is available should you need assistance or technical support. You can contact them by email at ideice@minerd.gob.do or by phone at 809-688-9700, extensions 7961 and 7962, Monday through Friday, from 9:00 a.m. to 4:00 p.m.

Would you like me to repeat the email address or phone number?

Everyone:

In the same email, you must download a Brochure and an Instruction Guide, which appear as download buttons at the end of the email. It is very important that you download both files and review them carefully. Note that both download buttons are at the end of the email.

We appreciate your cooperation and trust that the implementation of this program will be a success thanks to your efforts.

2nd Call (Only Reminders Group)

Everyone:

Hello, my name is XXX and I am calling on behalf of the Ministry of Education to give you an important message. Do you have a couple of minutes?

Are you currently in charge of the school XXX with Center Code XXX?

We are calling to remind you about the delivery of an email regarding the informational campaign Learning the Value of Education in the Dominican Republic (AVE-RD) on behalf of MINERD and IDEICE.

I first need to verify some of your information and information about your school. Are you XXX?

This email titled "School Calendar 2018-2019: AVE-RD Week" contains relevant information so that your school can implement the campaign.

This campaign consists of videos that provide information about the benefits of finishing school. These videos should be shown to secondary school students in your school between 25 and 28 September, as established in MINERD's 2018-2019 school calendar.

If you would like, we can resend this email to you. Would you like me to resend it?

Evidence Treatment Arm:

We would like to tell you that a study conducted in 2,500 schools in the Dominican Republic showed that there are proven benefits of implementing this campaign in your educational center.

Schools that showed these videos significantly improved their national exam scores by 0.6 points in total, and reduced their school dropout rates.

This study concludes that if all schools in the country showed these videos, the number of secondary school dropouts would decrease by 6,500 students per year.

Would you like me to repeat the information?

Incentive Treatment Arm:

If you implement the AVE-RD campaign in your school, you will have the opportunity to win a 7-inch Samsung tablet to help you manage your educational center. One out of every 10 people who participate in the raffle will win a tablet.

You will be able to participate in this raffle only if you implement the campaign in your school as established in the instructions in the email we sent you.

Would you like me to repeat the information?

Assistance Treatment Arm:

We would like to mention that a support center is available should you need assistance or technical support. You can contact them by email at ideice@minerd.gob.do or by phone at 809-688-9700, extensions 7961 and 7962, Monday through Friday, from 9:00 a.m. to 4:00 p.m.

Would you like me to repeat the email address or phone number?

Everyone:

In the same email, you must download a Brochure and an Instruction Guide, which appear as download buttons at the end of the email. It is very important that you download both files and review them carefully.

Note that both download buttons are at the end of the email.

We appreciate your cooperation and trust that the implementation of this program will be a success thanks to your efforts.

3rd Call (Only Reminders Group)

Hello, my name is XXX and I am calling on behalf of the Ministry of Education to give you an important message. Do you have a couple of minutes?

Are you currently in charge of the school XXX with Center Code XXX?

We are calling to remind you about the delivery of several emails regarding the

informational campaign Learning the Value of Education in the Dominican Republic (AVE-RD) on behalf of MINERD and IDEICE.

I first need to verify some of your information and information about your school. Are you XXX?

These emails contain relevant information so that your school can implement the campaign. This campaign consists of videos that provide information about the benefits of finishing school. These videos should be shown to secondary school students in your school starting on 25 September, as established in the MINERD 2018-2019 school calendar.

If you would like, we can resend this email to you. Would you like me to resend it? Please remember that you must send a WhatsApp message to the number XXX with photos of the video showings.

You must also complete the implementation form and send it to XXX. This form will be downloaded together with the videos.

We appreciate your cooperation and trust that the implementation of this program will be a success thanks to your efforts.

D Additional Results for Anti-Bullying Campaign

Here is the content of the messages we randomized in the follow-up experiment:

Motivation (all principals): *The school climate is an important factor that promotes learning. The school climate is something that can be managed, and the "Declaration for Good Treatment" can contribute to this end. Be part of the solution and let's help together to reduce bullying and bad treatment at school.*

Monitoring (for principals who implemented the AVE campaign): *According to our records, your school successfully implemented the Learning the Value of Education campaign (AVE-RD). We thank you for your participation, and we urge you to also participate in the "Declaration for Good Treatment". Again we will be monitoring compliance with this instruction, to ensure that this campaign is implemented properly.*

Monitoring (for principals who did not implement the AVE campaign): *According to our records, your school did not satisfactorily implement the previous campaign Learning the Value of Education (AVE-RD). We urge you to participate in the "Declaration for Good Treatment" on this occasion. Again we will be monitoring compliance with this instruction, to ensure that this campaign is implemented properly.*

A third intervention arm received additional reminders. While other principals were only contacted via email, the reminders treatment group received two additional calls from a call center. All three intervention arms were perfectly cross-randomized. Randomization was stratified based on whether the school was part of the evidence, incentive, reminders, or assistance treatment arm in the AVE intervention and based on whether the school had implemented the AVE campaign.

Table D.1 presents the balance check, Table D.2 shows the results of the multiple hypothesis correction, Tables D.3 and D.4 show the effects for each treatment cell, and Table D.5 shows the effects on student reports.

Table D.1: Balance Check for Anti-Bullying Campaign

	Control		Motivation		Monitoring		Reminders		
	Mean	St. D.	Coeff.	St. E.	Coeff.	St. E.	Coeff.	St. E.	N
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: School Characteristics									
Has Internet Connection	0.738	[0.440]	-0.004	(0.017)	0.013	(0.017)	0.000	(0.017)	2,717
Has Projector	0.753	[0.432]	0.014	(0.016)	-0.001	(0.016)	-0.007	(0.016)	2,719
Has Computer Lab	0.558	[0.497]	0.008	(0.019)	-0.005	(0.019)	0.014	(0.019)	2,717
Has Electricity	0.994	[0.077]	-0.007*	(0.004)	0.006	(0.004)	-0.006	(0.004)	2,719
Students in 7-12th Grade, 2018	188.632	[186.571]	-8.408	(8.458)	8.693	(8.472)	-7.335	(8.466)	2,724
Total PN Score 12th Grade, 2018	72.931	[7.263]	0.722*	(0.396)	-0.502	(0.393)	0.340	(0.396)	1,324
Total PN Score 8th Grade, 2016	69.418	[7.422]	-0.105	(0.337)	-0.098	(0.336)	0.183	(0.335)	1,925
Shares Building with Another Sample School	0.079	[0.271]	0.012	(0.011)	0.005	(0.011)	0.002	(0.011)	2,724
Offers Extended School Day	0.409	[0.492]	0.012	(0.019)	0.026	(0.019)	0.015	(0.019)	2,724
Located in Urban Area	0.737	[0.441]	-0.018	(0.019)	-0.013	(0.019)	-0.008	(0.019)	2,358
School Is Public	0.732	[0.443]	0.012	(0.016)	-0.013	(0.016)	-0.003	(0.016)	2,724
Treatment Group in AVE Pilot	0.332	[0.472]	0.026	(0.018)	-0.022	(0.018)	-0.014	(0.018)	2,724
Panel B: Principal Characteristics									
Experience of Principal	19.183	[8.045]	0.295	(0.333)	0.063	(0.334)	-0.471	(0.334)	2,111
Principal Has Post-Graduate Degree	0.550	[0.498]	-0.019	(0.020)	-0.017	(0.020)	-0.010	(0.020)	2,568
Principal Supervises Multiple Schools	0.083	[0.276]	-0.007	(0.011)	0.007	(0.011)	0.011	(0.011)	2,710
Principal Checks Emails Daily	0.592	[0.492]	0.023	(0.019)	-0.016	(0.019)	0.005	(0.019)	2,630
Believes Penalty after Non-Compliance is Very Likely	0.560	[0.497]	-0.002	(0.019)	0.028	(0.019)	0.003	(0.019)	2,723
Principal's Main Motivation is Professional Success	0.447	[0.498]	-0.007	(0.019)	0.022	(0.019)	-0.011	(0.019)	2,723
Believes Extra Info Would Have An Effect	0.868	[0.339]	-0.012	(0.013)	-0.010	(0.013)	0.013	(0.013)	2,724
Believes Students Undervalue Education	0.531	[0.500]	-0.023	(0.019)	-0.023	(0.019)	0.019	(0.019)	2,630
Believes Lack of Assistance is Major Barrier	0.376	[0.485]	0.001	(0.019)	-0.026	(0.019)	0.034*	(0.019)	2,724
Number of Valid Emails	1.347	[0.698]	-0.004	(0.026)	-0.025	(0.026)	0.039	(0.026)	2,724
Principal Has Multiple Positions	0.690	[0.463]	-0.014	(0.021)	-0.001	(0.021)	-0.024	(0.021)	1,994
Principal Has No Secretary	0.264	[0.442]	0.001	(0.022)	-0.012	(0.022)	0.012	(0.022)	1,499
Joint Test p-value			0.43		0.25		0.71		

Notes: This table shows balance for baseline covariates. Panel A reports on school-level outcomes and Panel B on principal-level outcomes. Differences in sample sizes across variables reflect missing data. Column (1) reports the control group mean of the dependent variable. Columns (3), (5), and (7) report the difference in the dependent variable from OLS regressions of each outcome on indicator variables for assignments to the motivation treatment group, the monitoring treatment group, and the reminders treatment group, and stratification dummies. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.2: Multiple Hypothesis Correction for Anti-Bullying Campaign

	Index Components			
	Primary Outcome	Secondary Outcomes		
	Submitted Photo	Clicked on Declaration	Submitted Form	Adoption Index (in Std)
	(1)	(2)	(3)	(4)
Monitoring	0.009	0.040	0.001	0.001
Motivation	0.207	0.105	0.381	0.158
Reminders	0.001	0.001	0.001	0.001

Notes: This table reports sharpened false discovery rate (FDR) q-values corresponding to the estimates in Table 4. Q-values are computed following Benjamini et al. (2006) and control the expected proportion of false discoveries among all rejected hypotheses. For each outcome column, the correction is applied across the family of three hypotheses corresponding to the three treatment arms.

Table D.3: Effects on Anti-Bullying Campaign Adoption by Treatment Cell

	Index Components			
	Primary Outcome	Secondary Outcomes		
	Submitted Photo	Clicked on Declaration	Submitted Form	Adoption Index (in Std)
	(1)	(2)	(3)	(4)
Motivation	0.026	-0.030	-0.003	-0.024
	(0.029)	(0.035)	(0.030)	(0.071)
Monitoring	0.043	0.065*	0.054*	0.149**
	(0.030)	(0.036)	(0.032)	(0.075)
Reminders	0.318***	0.342***	0.190***	0.764***
	(0.033)	(0.033)	(0.033)	(0.070)
Motivation + Monitoring	0.100***	0.071**	0.063**	0.200***
	(0.031)	(0.035)	(0.031)	(0.074)
Motivation + Reminders	0.217***	0.290***	0.158***	0.621***
	(0.032)	(0.035)	(0.033)	(0.070)
Monitoring + Reminders	0.310***	0.302***	0.231***	0.757***
	(0.033)	(0.034)	(0.034)	(0.073)
Monitoring + Motivation + Reminders	0.286***	0.307***	0.244***	0.760***
	(0.033)	(0.034)	(0.034)	(0.073)
Mean of Control Group	0.182	0.379	0.215	0.000
Observations	2724	2724	2724	2724

Notes: This table shows the treatment effects on different adoption measures of the anti-bullying campaign. We regress each outcome on indicator variables for each treatment cell, stratification dummies, and baseline controls selected by LASSO. Table D.1 shows the list of potential lasso controls. The adoption index (column 4) is standardized to have mean zero and standard deviation one in the control group. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.4: Pairwise Comparisons Across Treatment Cells in the Anti-Bullying Experiment

	Estimate/ Difference (1)	*p*-val (2)	Ex-post MDE (3)		Estimate/ Difference (4)	*p*-val (5)	Ex-post MDE (6)
Panel A: Pooled Treatment Effects (Table 4)							
Motivation	-0.011	0.514	0.048	Reminders	0.241	0.000***	0.049
Monitoring	0.045	0.008***	0.047				
Panel B: Cross-treatment Comparisons (Table 4)							
Motivation vs. Monitoring	-0.056	0.020**	0.067	Monitoring vs. Reminders	-0.196	0.000***	0.069
Motivation vs. Reminders	-0.252	0.000***	0.068				
Panel C: Treatment Cells vs. Control (Table D.3)							
Motivation Only	0.026	0.374	0.081	Mot+Rem	0.217	0.000***	0.090
Monitoring Only	0.043	0.159	0.085	Mon+Rem	0.310	0.000***	0.093
Reminders Only	0.318	0.000***	0.092	Mot+Mon+Rem	0.286	0.000***	0.093
Mot+Mon	0.100	0.001***	0.086				
Panel D: Pairwise Cross-cut Comparisons (Table D.3)							
Motivation Only vs. Monitoring Only	-0.017	0.591	0.088	Reminders Only vs. Mot+Mon	0.219	0.000***	0.097
Motivation Only vs. Reminders Only	-0.292	0.000***	0.094	Reminders Only vs. Mot+Rem	0.101	0.005***	0.100
Motivation Only vs. Mot+Mon	-0.074	0.020**	0.089	Reminders Only vs. Mon+Rem	0.008	0.832	0.103
Motivation Only vs. Mot+Rem	-0.191	0.000***	0.092	Reminders Only vs. Mot+Mon+Rem	0.032	0.378	0.103
Motivation Only vs. Mon+Rem	-0.285	0.000***	0.095	Mot+Mon vs. Mot+Rem	-0.117	0.001***	0.095
Motivation Only vs. Mot+Mon+Rem	-0.260	0.000***	0.095	Mot+Mon vs. Mon+Rem	-0.211	0.000***	0.099
Monitoring Only vs. Reminders Only	-0.276	0.000***	0.097	Mot+Mon vs. Mot+Mon+Rem	-0.186	0.000***	0.099
Monitoring Only vs. Mot+Mon	-0.057	0.083*	0.092	Mot+Rem vs. Mon+Rem	-0.094	0.010***	0.101
Monitoring Only vs. Mot+Rem	-0.174	0.000***	0.095	Mot+Rem vs. Mot+Mon+Rem	-0.069	0.057*	0.101
Monitoring Only vs. Mon+Rem	-0.268	0.000***	0.098	Mon+Rem vs. Mot+Mon+Rem	0.025	0.508	0.104
Monitoring Only vs. Mot+Mon+Rem	-0.243	0.000***	0.099				

Notes: This table reports pairwise comparisons of anti-bullying campaign adoption across all treatment cells based on the regression in column (1) in Tables 4 and D.3. The "Difference" column reports the difference in adoption rates between the two cells, and the "p-value" column the p-value from a test of their equality. The "Ex-post MDE" column reports the ex-post minimum detectable effect (80% power, 5% two-sided) for each comparison, in the same units as the "Difference" column. Treatment cells are abbreviated Mot (Motivation), Mon (Monitoring), and Rem (Reminders). Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table D.5: Effects on Student Reports for Anti-Bullying Experiment

	Student Share (1)	Student Share $\geq$ 50% (2)
Motivation	-0.012	-0.027
	(0.012)	(0.022)
Monitoring	0.014	0.018
	(0.012)	(0.022)
Reminders	0.084***	0.117***
	(0.012)	(0.023)
Mean of Control Group	0.526	0.545
Observations	1588	1588

Notes: This table reports effects on anti-bullying campaign adoption based on student reports. In Column (1), the outcome is the share of students who report either participating in the anti-bullying campaign themselves or observing participation by other students in their school. In Column (2), the outcome is an indicator equal to one if this share is at least 50%. We regress each outcome on indicator variables for assignments to the motivation treatment group, the monitoring treatment group, and the reminders treatment group, stratification dummies, and baseline controls selected by LASSO. Table D.1 shows the list of potential lasso controls. All regressions are estimated by OLS and robust standard errors are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

E Call Scripts for the Anti-Bullying Campaign

1st and 2nd Reminders Call

Everyone:

Hello, my name is XXX and I am calling on behalf of the Ministry of Education to give you an important message. Do you have a couple of minutes? Are you currently in charge of the school XXX with Center Code XXX?

We are calling to notify you about the delivery of an email regarding the Culture of Peace and Declaration of Good Treatment campaign sent on November 8. I first need to verify some of your information and information about your school. Are you XXX?

This email was sent by the Ministry of Education and IDEICE on behalf of Minerva Perez, Director of the Directorate of Guidance and Psychology, on Thursday, November 8, and contains relevant information so that your school can implement the campaign.

I wanted to notify you that we sent the email to the following accounts: XXX

Have you been able to see this email?

The MINERD 2018-2019 school calendar establishes the month of November as the month of good treatment in schools. For this reason, MINERD, through the Directorate of Guidance and Psychology, is promoting the Culture of Peace and the Declaration of Good Treatment campaign. This campaign consists of all classrooms collaboratively creating a mural or bulletin board related to good treatment on November 23, and reading and signing the "Declaration for Good Treatment."

Monitoring - Implemented AVE Campaign:

According to our records, your educational center successfully implemented the Learning the Value of Education (AVE-RD) campaign.

We appreciate your participation, and we encourage you to also participate in the "Declaration for Good Treatment." MINERD and IDEICE will again monitor com-

pliance with this instruction to ensure that this campaign is properly implemented.

Monitoring - Did Not Implement AVE Campaign:

According to our records, your educational center did not successfully implement the previous campaign Learning the Value of Education (AVE-RD).

We encourage you to participate in the "Declaration for Good Treatment" this time. MINERD and IDEICE will again monitor compliance with this instruction to ensure that this campaign is properly implemented.

Motivation:

School climate is an important factor that promotes learning.

Moreover, school climate is something that can be managed, and the "Declaration for Good Treatment" can contribute to this goal.

We invite you to be part of the solution so that together we can help reduce bullying and improve treatment in schools.

Everyone:

To comply with this instruction, you must follow the steps below:

Step 1: Download the "Declaration for Good Treatment" that you will find in the email and print one copy for each student. If you cannot print one for each student, print one per classroom.

Step 2: Download the Implementation Guide that you will find in the email and follow the instructions to carry out the campaign on November 23 in all secondary classrooms.

Step 3: Document compliance with the declaration by sending photos via WhatsApp on November 23 as indicated in the guide (one photo per classroom), and complete an implementation form that you must send to XXX before November 30.

On the webpage for the month of good treatment, which is included in the email we sent you, you will find summarized instructions, additional materials on the appli-

cation protocols for the Culture of Peace, and contact information should you have any questions about the activities related to the campaign.

Please review the email we sent and download the attached files using the download buttons as soon as you have a moment. It is very important that you download both files and review them carefully.

We appreciate your cooperation, and we trust that the implementation of this program will be a success thanks to your efforts.

F Downstream Outcomes

The original AVE trial reduced dropout and raised test scores (Berry et al., 2018). This appendix estimates the effect of program adoption on these two outcomes in the scale-up.

Data. We link students to administrative enrollment and test-score data through their national student identifier. We start from the 2018–2019 enrollment roster of the schools in our study, which covers 2,596 of the 2,724 schools.[25] For each student, we observe enrollment in any school in 2019–2020 from the national enrollment roster, which covers public, private, and semi-public schools. We also observe scores on the *Pruebas Nacionales* (PN), the national exam at the end of secondary school, through 2025.

Outcomes. Dropout equals one if a student enrolled in grades 7 to 11 in 2018–2019 does not appear anywhere in the 2019–2020 national roster. We exclude grade 12 students, who finish secondary school that year. Test scores are the average PN score across Spanish, Mathematics, Social Science, and Natural Science. We use the first sitting in the student's scheduled exam year, the year the student would reach grade 12, and require scores in all four subjects. We standardize scores by the mean and standard deviation of control group students. The PN was not held in 2020 and 2021, so students in grades 10 and 11 in 2018–2019 have no score in their scheduled year. The dropout sample has 485,326 students in 2,593 schools, and the test-score sample has 237,089 students in 2,591 schools. Table F.1 shows both samples by grade.

Specification. We measure program adoption by photo submission by October 11, the primary outcome in Table 3. We instrument it with assignment to the reminders treatment. Table 3 shows that the evidence, incentive, and assistance treatments had no effect on photo submission, so using them as instruments would add weak instruments without adding identifying variation. We include them as

[25] Of the 128 schools not covered, 114 had no students in grades 7 to 12 in 2018–2019, mostly former grade 7–8 schools that stopped offering these grades after the reorganization of secondary education, and 14 do not appear under their school identifier. Coverage is not significantly related to the evidence treatment ($p = 0.25$), and schools assigned to reminders are 2.2 percentage points more likely to be covered ($p = 0.01$); a joint test of all four treatments gives $p = 0.10$. Because the grades each school taught in 2018–2019 were set before the experiment began, these differences reflect chance rather than the treatments.

controls instead, since they were assigned together with reminders in the same cross-randomized design. The regressions also control for all potential lasso controls in Table A.3, including the school's 2018 grade-12 and 2016 grade-8 PN scores, and grade and randomization-stratum fixed effects. We weight each student by the inverse of the school's sample size, so each school receives equal weight, and cluster standard errors by school. The estimates are local average treatment effects for schools whose adoption responds to reminders.

Results. Table F.2 shows the results. The first stage is strong: the reminders treatment raises photo submission by 21 percentage points (s.e. 2.2), close to the 20.3 percentage points in Table 3, with a first-stage F statistic of about 88. The reduced-form effects of reminders go in the expected direction but are imprecise: dropout falls by 0.10 percentage points (s.e. 0.33) and test scores rise by 0.006 standard deviations (s.e. 0.013). The same holds for the two-stage least squares estimates. Program adoption lowers dropout by 0.50 percentage points (s.e. 1.57), from a control mean of 8.4%, and raises test scores by 0.028 standard deviations (s.e. 0.063). Reminders do not change the share of students who take the exam on schedule, which falls by 0.30 percentage points (s.e. 0.56) from a control mean of 60%. Thus, selection into the test-score sample is similar in schools with and without reminders.

Interpretation. These estimates are too noisy to distinguish a zero effect from the intent-to-treat effects of the original trial, a 3 percentage point fall in dropout and a 0.05 standard deviation gain in test scores. Our 95% confidence intervals, $[-3.57, 2.58]$ percentage points for dropout and $[-0.10, 0.15]$ standard deviations for test scores, contain both zero and these effects.[26] The estimates are imprecise because reminders, our only source of identifying variation in adoption, raise it by 21 percentage points. At 80% power and a 5% significance level, the minimum detectable effects of adoption are 4.4 percentage points on dropout and 0.18 standard deviations on test scores.

[26]Implementation in the original trial was 83%, so its effects of implementation are about -3.6 percentage points on dropout and $+0.06$ standard deviations on test scores. Our test-score interval contains this effect, and our dropout interval narrowly excludes it, although the two estimates apply to different schools.

Table F.1: Sample by Grade

Grade in 2018–2019	Scheduled PN year	Students	Dropped out next year (%)	Students with PN score	Mean PN score
7	2024	102,876	5.7	54,149	57.7
8	2023	97,751	7.8	54,251	57.3
9	2022	99,353	9.4	58,484	60.3
10	2021	98,624	9.2	—	—
11	2020	86,722	7.6	—	—
12	2019	73,618	—	70,205	62.0
Dropout sample (grades 7–11)		485,326	7.9		
Test-score sample (grades 7, 8, 9, 12)				237,089	59.5

Notes: Students enrolled in the schools in our study in 2018–2019, by grade. Scheduled PN year: the year the student would reach grade 12 without repeating a grade; the PN was not held in 2020 and 2021. Dropped out next year: share of students not enrolled anywhere in 2019–2020; grade 12 students are excluded. Students with PN score: students with scores in all four subjects at the first sitting of their scheduled exam. Mean PN score: average raw score across the four subjects, in points.

Table F.2: Effects of Program Adoption on Dropout and Test Scores

	(1) Dropped out next year (pp)	(2) PN score, scheduled year, first sitting (SD)
Panel A: 2SLS		
Submitted photo	-0.50	0.028
	(1.57)	(0.063)
95% CI	[-3.57, 2.58]	[-0.096, 0.152]
Panel B: First stage (dependent variable: submitted photo)		
Reminders	0.210***	0.208***
	(0.022)	(0.022)
First-stage F	89.0	87.4
Panel C: Reduced form		
Reminders	-0.10	0.006
	(0.33)	(0.013)
Control mean (%; raw points)	8.40	57.38
Schools	2,593	2,591
Students	485,326	237,089

Notes: Panel A reports two-stage least squares estimates of the effect of program adoption, measured by photo submission by October 11 (the primary outcome in Table 3), instrumented by assignment to the reminders treatment. Panel B reports the first stage, with coefficients as shares, and Panel C the reduced form. All regressions include the evidence, incentive, and assistance treatment indicators, all potential lasso controls in Table A.3, and grade and randomization-stratum fixed effects. Students are weighted by the inverse of their school's sample size, so each school receives equal weight. Standard errors clustered by school are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.